

Preface

At the root of measure theory is a rigorous treatment of how to extend the notions of length (on the real line), area (on the Euclidean plane), and volume (in three-space) to a larger collection of sets than those seen in, say, courses on Calculus. We have seen in a first course in analysis that there exist sets with some exceptionally strange properties (the Cantor set being a classic example). We would like to be able to discuss these types of sets in the context of measuring their sizes (be it a length, area, volume, etc). While the historical impetus was to solve this problem for sets in $\mathbb{R}^n$, one finds that it is not hard to abstract the theory to more general sets. Accordingly we take up this abstract study. This is not just for the sake of abstraction. In fact one can find applications for abstract measure theory in, for example, probability theory (where the measure of a set corresponds to the likelihood of an element in that set being chosen under some random process) and physics (where one might want the measure of a set to correspond to the mass of some object occupying that space).

Contents

Contents	i
1 Measures	1
1.1 σ -Algebras	2
1.2 Measures	6
1.3 Outer Measures	10
1.4 Measures on $\mathbb{R}$	14
2 Integration	22
2.1 Measurable Functions	22
2.2 Integration	24
2.3 Lebesgue Integration on $\mathbb{R}$	37
2.4 Modes of Convergence	40
2.5 Integration in $\mathbb{R}^n$	42
3 Differentiation	43
3.1 Signed Measures	43

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Chapter 1

Measures

Ideally we would like to have a function μ that assigns each $E \subset \mathbb{R}$ to a number $\mu(E) \in [0, \infty]$. Such a function μ should possess the following properties:

1. The unit interval should have a measure equal to one—that is, $\mu([0, 1]) = 1$;
2. If E and F are congruent subsets of $\mathbb{R}$ then $\mu(E) = \mu(F)$;
3. If $E_1, E_2, \dots$ are pairwise disjoint, then $\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu(E_j)$.

Unfortunately, a function possessing all of these traits does not exist.

Theorem 1.0.1. *A function $\mu : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ satisfying (1), (2), and (3) from above does not exist.*

Proof. Suppose such a function μ does exist. By (1) we have $\mu([0, 1]) = 1$. Define a relation on $[0, 1]$ as follows: $x \sim y$ if and only if $x - y \in \mathbb{Q}$. This is an equivalence relation. Using the Axiom of Choice, let N be a set which contains exactly one element from each equivalence class. Now for each $q \in \mathbb{Q} \cap [0, 1)$, define:

$$N_q = \{x + q \mid x \in N \cap [0, 1 - q)\} \cup \{x + (q - 1) \mid x \in N \cap [1 - q, 1)\}.$$

That is, the points in N_q are obtained by "rotating" the points in N by a rational number q . Clearly $N_q \subset [0, 1)$, hence $\bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q \subset [0, 1)$. We have the following three properties:

- (a) For any $q \in \mathbb{Q}$, $\mu(N) = \mu(N_q)$
- (b) If $r, s \in \mathbb{Q}$, $r \neq s$, then $N_r \cap N_s = \emptyset$.
- (c) For every $y \in [0, 1)$, there exists a $q \in \mathbb{Q}$ such that $y \in N_q$.

Property (a) follows from the fact that N and N_r are congruent, hence:

$$\begin{aligned} \mu(N) &= \mu(N \cap [0, 1 - q)) + \mu(N \cap [1 - q, 1)) \\ &= \mu(\{x + q \mid x \in N \cap [0, 1 - q)\}) + \mu(\{x + q - 1 \mid x \in N \cap [1 - q, 1)\}) \quad \text{"Rotation"} \\ &= \mu(N_r). \end{aligned}$$

Property (b) follows by contradiction—if $x \in N_r \cap N_s$, then $x - r$ and $x - s$ would be distinct elements of N . However, $(x - r) - (x - s) = s - r \in \mathbb{Q}$, so $x - r$ and $x - s$ are in the same equivalence class. But this contradicts the way which we've constructed N . For property (c), given $y \in [0, 1)$, let $x \in N$ satisfy $x \sim y$. Since x and y are in the same equivalence class, we can find $q \in \mathbb{Q}$ such that:

$$y = \begin{cases} x + q, & x < y \\ x - (q - 1) & x > y \end{cases}.$$

Hence $y \in N_q$, giving $[0, 1) \subset N_q \subset \bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q$. We've obtained the equality:

$$\bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q = [0, 1).$$

Property (b) says that this is a disjoint union:

$$\bigsqcup_{q \in \mathbb{Q} \cap [0, 1)} N_q = [0, 1).$$

Finally, with property (a), we have:

$$\begin{aligned} 1 &= \mu([0, 1)) \\ &= \mu\left(\bigsqcup_{q \in \mathbb{Q} \cap [0, 1)} N_q\right) \\ &= \sum_{q \in \mathbb{Q} \cap [0, 1)} \mu(N_q) \\ &= \sum_{n=1}^{\infty} \mu(N). \end{aligned}$$

This is impossible, hence there does not exist a μ satisfying (1), (2), and (3). $\square$

A good guess for this failure would be that we assumed countable additivity as opposed to finite additivity. This is not the case. In dimensions $n \geq 3$, assuming such a weak form of μ leads to the following result:

Theorem 1.0.2 (Banach-Tarski). *Let U and V be arbitrary bounded open sets in $\mathbb{R}^n$ for $n \geq 3$. There exist $k \in \mathbb{N}$ and subsets $E_1, \dots, E_k, F_1, \dots, F_k$ of $\mathbb{R}^n$ such that:*

- *the E_j 's are disjoint and their union is U ;*
- *the F_j 's are disjoint and their union is V ;*
- *E_j is congruent to F_j for all $1 \leq j \leq k$.*

Ultimately, the problem comes from the fact we defined μ on *every possible* subset of $\mathbb{R}$. We can't hope to do this, since as we showed there exists bizarre sets like N which cannot be measured. Instead, we will define μ on a special class of subsets of $\mathbb{R}^n$.

§ 1.1. σ -Algebras

Definition 1.1.1. *Let X be a set. An algebra of sets $\mathcal{A} \subset \mathcal{P}(X)$ is a collection of subsets of X which is:*

- (1) *closed under finite union: if $E_1, E_2, \dots, E_n \in \mathcal{A}$, then $\bigcup_{i=1}^n E_i \in \mathcal{A}$;*
- (2) *closed under complement: if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.*

If $\mathcal{A}$ is closed under countable union, then we say $\mathcal{A}$ is a σ -algebra.

Proposition 1.1.1. *Let $\mathcal{A}$ be an algebra of sets.*

- (1) $\mathcal{A}$ is closed under finite intersection.
- (2) $\mathcal{A}$ is closed under set difference.
- (3) $\emptyset \in \mathcal{A}$ and $X \in \mathcal{A}$.

Proof. (1) Let $E_1, E_2, \dots, E_n \in \mathcal{A}$. Then $\bigcap_{i=1}^n E_i = \left(\bigcup_{i=1}^n E_i^c\right)^c \in \mathcal{A}$. (2) If $E_1, E_2 \in \mathcal{A}$, then $E_1 \setminus E_2 = E_1 \cap E_2^c \in \mathcal{A}$. (3) For any $E \in \mathcal{A}$, we have $\emptyset = E \cap E^c \in \mathcal{A}$ and $X = E \cup E^c \in \mathcal{A}$. $\square$

Proposition 1.1.2. *Let $\mathcal{A}$ be an algebra of sets. The following are equivalent:*

- (1) $\mathcal{A}$ is a σ -algebra.
- (2) $\mathcal{A}$ is closed under countable disjoint union.

Proof. If $\mathcal{A}$ is a σ -algebra, then it is closed under countable union, hence it must be closed under countable disjoint union. Conversely, suppose $\mathcal{A}$ is closed under countable disjoint union. Let $\{E_n\}_{n=1}^\infty$ be a family of sets in $\mathcal{A}$. Define:

$$\begin{aligned} F_1 &= E_1, \\ F_2 &= E_2 \setminus E_1 \\ F_3 &= E_3 \setminus (E_1 \cup E_2) \\ &\vdots \\ F_n &= E_n \setminus (E_1 \cup E_2 \cup \dots \cup E_{n-1}). \end{aligned}$$

Let $i < j$ be natural numbers and consider $F_i \cap F_j$. If $x \in F_i$, then $x \in E_i$. Since $i < j$, we have $x \in E_1 \cup \dots \cup E_{j-1}$, hence $x \notin F_j$. Thus $F_i \cap F_j = \emptyset$. Inductively, $\{F_n\}_{n=1}^\infty$ is a countable family of disjoint sets.

Clearly $\bigcup_{n=1}^\infty F_n \subset \bigcup_{n=1}^\infty E_n$. Conversely, let $x \in \bigcup_{n=1}^\infty E_n$. Find the smallest natural number j such that $x \in E_j$. Then $x \notin E_i$ for $i < j$. Hence $x \in E_j \setminus (E_1 \cup \dots \cup E_{j-1}) = F_j \subset \bigcup_{n=1}^\infty F_n$.

Since $\mathcal{A}$ is closed under countable disjoint unions, we have $\bigcup_{n=1}^\infty E_n = \bigcup_{n=1}^\infty F_n \in \mathcal{A}$. Since $\{E_n\}_{n=1}^\infty$ was an arbitrary family of sets, this means $\mathcal{A}$ is a σ -algebra. $\square$

Proposition 1.1.3. *If $\{\mathcal{E}_i\}_{i \in I}$ is a family of σ -algebras, then $\bigcap_{i \in I} \mathcal{E}_i$ is a σ -algebra.*

Proof. Let $\{E_n\}_{n=1}^\infty$ be a family of sets contained in $\bigcap_{i \in I} \mathcal{E}_i$. Then each $\{E_n\}_{n=1}^\infty$ is contained in $\mathcal{E}_i$. Since $\mathcal{E}_i$ is a σ -algebra for every $i \in I$, we have $\bigcup_{n=1}^\infty E_n \in \mathcal{E}_i$ for every $i \in I$, which means $\bigcup_{n=1}^\infty E_n \in \bigcap_{i \in I} \mathcal{E}_i$.

Similarly, let $F \in \bigcap_{i \in I} \mathcal{E}_i$. Then $F \in \mathcal{E}_i$ for every $i \in I$. Since $\mathcal{E}_i$ is a σ -algebra for every $i \in I$, we have $F^c \in \mathcal{E}_i$ for every $i \in I$, which means $F^c \in \bigcap_{i \in I} \mathcal{E}_i$. Thus $\bigcap_{i \in I} \mathcal{E}_i$ is a σ -algebra. $\square$

Definition 1.1.2. *Let X be a set and $\mathcal{E} \subset \mathcal{P}(X)$. The σ -algebra generated by $\mathcal{E}$ is $\mathfrak{M}(\mathcal{E}) := \bigcap \{\mathcal{A} \mid \mathcal{E} \subset \mathcal{A}, \mathcal{A} \text{ is a } \sigma\text{-algebra}\}$.*

Lemma 1.1.1. *If $\mathcal{E} \subset \mathfrak{M}(\mathcal{F})$, then $\mathfrak{M}(\mathcal{E}) \subset \mathfrak{M}(\mathcal{F})$.*

Proof. Since $\mathfrak{M}(\mathcal{F})$ is a σ -algebra containing $\mathcal{E}$, it must contain $\mathfrak{M}(\mathcal{E})$, as $\mathfrak{M}(\mathcal{E})$ is the smallest σ -algebra generated by $\mathcal{E}$. $\square$

Given a metric space X , we might only be interested in what is generated by its open sets.

Definition 1.1.3. *Let X be a metric space and τ_X the set containing all of its open sets. The Borel σ -algebra on X is the set $\mathcal{B}_X := \mathfrak{M}(\tau_X)$. Members of $\mathcal{B}_X$ are called Borel sets.*

Talk about G delta and F sigma sets.

Proposition 1.1.4. $\mathcal{B}_{\mathbb{R}}$ is generated by each of the following:

- (1) the open intervals: $\mathcal{E}_1 = \{(a, b) \mid a < b\}$,
- (2) the closed intervals: $\mathcal{E}_2 = \{[a, b] \mid a < b\}$,
- (3) the half-open intervals: $\mathcal{E}_3 = \{(a, b] \mid a < b\}$ or $\mathcal{E}_4 = \{[a, b) \mid a < b\}$,
- (4) the open rays: $\mathcal{E}_5 = \{(a, \infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) \mid a \in \mathbb{R}\}$,
- (5) the closed rays: $\mathcal{E}_7 = \{(a, \infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a) \mid a \in \mathbb{R}\}$.

Proof. We are required to show for each $\mathcal{E}_i$ that $\mathfrak{M}(\mathcal{E}_i) = \mathcal{B}_{\mathbb{R}}$.

(1) Let $I \in \mathcal{E}_1$ be an open interval. Then $I \in \mathcal{B}_{\mathbb{R}}$ because I is an open subset of $\mathbb{R}$. From this we have $\mathcal{E}_1 \subset \mathcal{B}_{\mathbb{R}}$, but because $\mathcal{B}_{\mathbb{R}}$ is a σ -algebra containing $\mathcal{E}_1$, and $\mathfrak{M}(\mathcal{E}_1)$ is the smallest σ -algebra containing $\mathcal{E}_1$, we have $\mathfrak{M}(\mathcal{E}_1) \subset \mathcal{B}_{\mathbb{R}}$. Conversely, let $U \in \tau_{\mathbb{R}}$. Then U is at most the countable union of disjoint open intervals, hence $U \in \mathfrak{M}(\mathcal{E}_1)$. From this we have $\tau_{\mathbb{R}} \subset \mathfrak{M}(\mathcal{E}_1)$, but because $\mathfrak{M}(\mathcal{E}_1)$ is a σ -algebra containing $\tau_{\mathbb{R}}$, and $\mathfrak{M}(\tau_{\mathbb{R}})$ is the smallest σ -algebra containing $\tau_{\mathbb{R}}$, we have $\mathfrak{M}(\tau_{\mathbb{R}}) = \mathcal{B}_{\mathbb{R}} \subset \mathfrak{M}(\mathcal{E}_1)$.

(2) Let $I \in \mathcal{E}_1$ be an open interval. Then I^c is a closed interval, hence it is contained in $\mathfrak{M}(\mathcal{E}_2)$. Whence $\mathfrak{M}(\mathcal{E}_1) \subset \mathfrak{M}(\mathcal{E}_2)$. Conversely, let $C \in \mathcal{E}_2$ be a closed interval. Then C can be written as the countable intersection of open sets. This means $C \in \mathfrak{M}(\mathcal{E}_1)$. Whence $\mathfrak{M}(\mathcal{E}_2) \subset \mathfrak{M}(\mathcal{E}_1)$. Thus $\mathfrak{M}(\mathcal{E}_2) = \mathfrak{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}$ $\square$

Given two sets X_1 and X_2 with respective σ -algebras $\mathcal{M}_1$ and $\mathcal{M}_2$, we'd like to better understand what a σ -algebra on $X_1 \times X_2$ looks like.

Definition 1.1.4. Let $\{X_{\alpha}\}_{\alpha \in I}$ be an indexed family of nonempty sets, $X = \prod_{\alpha \in I} X_{\alpha}$, and $\pi_{\alpha} : X \rightarrow X_{\alpha}$ the projection maps. If $\mathcal{M}_{\alpha}$ is a σ -algebra on X_{α} for each α , the product σ -algebra on X is:

$$\bigotimes_{\alpha \in I} \mathcal{M}_{\alpha} := \mathfrak{M}\left(\{\pi_{\alpha}^{-1}(E_{\alpha}) \mid E_{\alpha} \in \mathcal{M}_{\alpha}, \alpha \in I\}\right)$$

The significance of this definition will become clear later. If X_1 and X_2 are sets with σ -algebras $\mathcal{M}_1$ and $\mathcal{M}_2$ respectively, one might wager a more intuitive definition for the σ -algebra on $X_1 \times X_2$ to be:

$$\mathfrak{M}\left(\{E_1 \times E_2 \mid E_1 \in \mathcal{M}_1, E_2 \in \mathcal{M}_2\}\right).$$

In fact, these two definitions are equivalent if we're only taking the cartesian product of countably many sets! Before proving this, note that if $E_i \subset X_i$ and $\pi_i : X_1 \times X_2 \rightarrow X_i$ for $i = 1, 2$, then:

$$\begin{aligned} \pi_1^{-1}(E_1) &= E_1 \times X_2, \\ \pi_2^{-1}(E_2) &= X_1 \times E_2. \end{aligned}$$

We can see that:

$$\begin{aligned} \pi_1^{-1}(E_1) \cap \pi_2^{-1}(E_2) &= (E_1 \times X_2) \cap (X_1 \times E_2) \\ &= \{(e_1, x_2) \mid e_1 \in E_1, x_2 \in X_2\} \cap \{(x_1, e_2) \mid x_1 \in X_1, e_2 \in E_2\} \\ &= \{(x, y) \mid (x \in E_1 \wedge x \in X_1) \wedge (y \in X_2 \wedge y \in E_2)\} \\ &= \{(x, y) \mid x \in E_1 \wedge y \in E_2\} \\ &= E_1 \times E_2. \end{aligned}$$

Proposition 1.1.5. *Let A be a countable indexing set, $\{X_\alpha\}_{\alpha \in A}$ an indexed family of nonempty sets, $X = \prod_{\alpha \in A} X_\alpha$, $\pi_\alpha : X \rightarrow X_\alpha$ the projection maps, and $\mathcal{M}_\alpha$ a σ -algebra on X_α for each $\alpha \in A$. Then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:*

$$\left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha \right\}$$

Proof. Denote $\mathcal{O}$ as the σ -algebra generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$. If $E_\alpha \in \mathcal{M}_\alpha$, then $\pi_\alpha^{-1}(E_\alpha) = \prod_{\beta \in A} E_\beta$, where $E_\beta = X_\beta$ for $\beta \neq \alpha$. This gives the inclusion:

$$\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\} \subset \mathcal{O}.$$

But since $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is the smallest σ -algebra generated by $\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\}$, we get:

$$\bigotimes_{\alpha \in A} \mathcal{M}_\alpha \subset \mathcal{O}.$$

Conversely, note that $\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha)$. Since each $\pi_\alpha^{-1}(E_\alpha)$ is in $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$, and since $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is closed under countable intersections, we have that $\prod_{\alpha \in A} E_\alpha \in \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Hence we have the following inclusion:

$$\left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha \right\} \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha.$$

But since $\mathcal{O}$ is the smallest σ -algebra generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$, it must be the case that $\mathcal{O} \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Since we showed inclusion in both directions, $\mathcal{O} = \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$; i.e., $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$. $\square$

If there exists a set $\mathcal{E}_\alpha$ such that $\mathcal{M}_\alpha = \mathfrak{M}(\mathcal{E}_\alpha)$, then we can generate the product similar algebra in a different way.

Lemma 1.1.2. **** Let $\{X_\alpha\}_{\alpha \in A}$ be a family of nonempty sets, $\mathcal{E}_\alpha \subset \mathcal{P}(X_\alpha)$, and $\mathcal{M}_\alpha := \mathfrak{M}(\mathcal{E}_\alpha)$ for each $\alpha \in A$. Then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:*

$$\mathcal{F}_1 = \{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\}.$$

Proof. Automatically we have $\mathcal{F}_1 \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$, hence $\mathfrak{M}(\mathcal{F}_1) \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Conversely, consider the set $S := \{E \subset X_\alpha \mid \pi_\alpha^{-1}(E) \in \mathfrak{M}(\mathcal{F}_1)\}$. This set is a σ -algebra. Indeed, if $\{E_n\}_{n \in \mathbb{N}}$ is a family of elements of S , then $\{\pi_\alpha^{-1}(E_n)\}_{n \in \mathbb{N}}$ is a family of elements of $\mathfrak{M}(\mathcal{F}_1)$. Since $\mathfrak{M}(\mathcal{F}_1)$ is a σ -algebra, we get:

$$\bigcup_{n \in \mathbb{N}} \pi_\alpha^{-1}(E_n) = \pi_\alpha^{-1}\left(\bigcup_{n \in \mathbb{N}} E_n\right).$$

Hence $\bigcup_{n \in \mathbb{N}} E_n \in S$. A similar argument can be used to show S is closed under complements. With this in mind, note that we also have $\mathcal{E}_\alpha \subset \mathfrak{M}(\mathcal{F}_1)$. Indeed, for any $E \in \mathcal{E}_\alpha$, **would like to finish some day...** $\square$

This result helps us prove an interesting fact about Borel sets.

Corollary 1.1.1. *** Let $X_1, \dots, X_n$ be metric spaces and let $X = \prod_{j=1}^n X_j$, equipped with the product metric. Then $\bigotimes_{j=1}^n \mathcal{B}_{X_j} \subset \mathcal{B}_X$. If the X_j 's are separable, then $\bigotimes_{j=1}^n \mathcal{B}_{X_j} = \mathcal{B}_X$.

Proof. would like to finish some day... □

Corollary 1.1.2. *** $\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{j=1}^n \mathcal{B}_{\mathbb{R}}$.

Proof. would like to finish some day... □

§ 1.2. Measures

Definition 1.2.1. Let X be a nonempty set and $\mathcal{M}$ a σ -algebra on X . A measure is a function:

$$\mu : \mathcal{M} \rightarrow [0, \infty]$$

satisfying:

(1) $\mu(\emptyset) = 0$;

(2) if $\{E_j\}_{j=1}^\infty \subset \mathcal{M}$ are pairwise disjoint, and then $\mu\left(\bigcup_{j=1}^\infty E_j\right) = \sum_{j=1}^\infty \mu(E_j)$.

We call the ordered triple $(X, \mathcal{M}, \mu)$ a measure space. The sets $E \subset \mathcal{M}$ are called measurable sets. If $\mu(X) < \infty$, then μ is called finite. If $X = \bigcup_{j=1}^\infty E_j$, $E_j \in \mathcal{M}$ for all j , and $\mu(E_j) < \infty$, then μ is σ -finite.

Example 1.2.1.

(1) Let X be a nonempty set. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by $\mu(E) = \sum_{x \in E} 1$ (note that this series exists as our codomain is the extended real numbers). Then $(X, \mathcal{P}(X), \mu)$ is a measure space, and we call μ the counting measure.

(2) Let X be nonempty and fix a point $x_0 \in X$. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by:

$$\mu(E) = \begin{cases} 0, & x_0 \notin E \\ 1, & x_0 \in E. \end{cases}$$

Then $(X, \mathcal{M}, \mu)$ is a measure space, and we call μ the dirac measure.

(3) Examples (1) and (2) can be generalized. Let X be a nonempty set and $f : X \rightarrow [0, \infty]$. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by $\mu(E) = \sum_{x \in E} f(x)$. Then $(X, \mathcal{P}(X), \mu)$ is a measure space. If $f : X \rightarrow [0, \infty]$ is the identity function, then μ is the counting measure. If $f : X \rightarrow [0, \infty]$ is the indicator function $\mathbb{1}_{\{x_0\}}$, then μ is the dirac measure.

(4) Let X be an uncountable set and let $\mathcal{M}$ be the σ -algebra of countable or co-countable sets. Define $\mu : \mathcal{M} \rightarrow [0, \infty]$ by:

$$\mu(E) = \begin{cases} 0, & E \text{ is countable} \\ 1, & E \text{ is co-countable.} \end{cases}$$

Then $(X, \mathcal{M}, \mu)$ is a measure space.

Proposition 1.2.1. Let $(X, \mathcal{M}, \mu)$ be a measure space and $E, F \in \mathcal{M}$. Then $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$.

Proof. (1) We can write $E \cup F = (E \setminus F) \sqcup (E \cap F) \sqcup (F \setminus E)$. This means $\mu(E \cup F) = \mu(E \setminus F) + \mu(E \cap F) + \mu(F \setminus E)$. This is equivalent to:

$$\begin{aligned}\mu(E \cup F) &= \left(\mu(E \setminus F) + \mu(E \cap F) \right) + \left(\mu(F \setminus E) + \mu(E \cap F) \right) - \mu(E \cap F) \\ &= \mu\left((E \setminus F) \cup (E \cap F)\right) + \mu\left((F \setminus E) \cup (E \cap F)\right) - \mu(E \cap F) \\ &= \mu(E) + \mu(F) - \mu(E \cap F)\end{aligned}$$

Hence $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$. □

Proposition 1.2.2. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

- (1) *If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$ (Monotonicity).*
- (2) *Let $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$. Then $\mu\left(\bigcup_{i=1}^\infty E_i\right) \leq \sum_{i=1}^\infty \mu(E_i)$ (Subadditivity).*
- (3) *Let $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$ with $E_1 \subset E_2 \subset E_3 \subset \dots$. Then $\mu\left(\bigcup_{i=1}^\infty E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i)$ (Continuity from below).*
- (4) *Let $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$ with $E_1 \supset E_2 \supset E_3 \supset \dots$ provided $\mu(E_1) < \infty$. Then $\mu\left(\bigcap_{i=1}^\infty E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i)$*

Proof. (1) We can write $F = E \sqcup (F \setminus E)$, giving $\mu(F) = \mu(E \sqcup (F \setminus E)) = \mu(E) + \mu(F \setminus E) \geq \mu(E)$.

(2) Given $\{E_i\}_{i=1}^\infty \subset \mathcal{M}$, define $F_i = E_i \setminus \bigcup_{k=1}^{i-1} E_k$ for each i . Then $\{F_i\}_{i=1}^\infty$ is a sequence of pairwise disjoint subsets of X . In particular, we have $F_i \subset E_i$ for each i , and $\bigcup_{i=1}^\infty E_i = \bigcup_{i=1}^\infty F_i$. Together:

$$\begin{aligned}\mu\left(\bigcup_{i=1}^\infty E_i\right) &= \mu\left(\bigcup_{i=1}^\infty F_i\right) \\ &= \sum_{i=1}^\infty \mu(F_i) \\ &\leq \sum_{i=1}^\infty \mu(E_i).\end{aligned}$$

(3) Define $E_0 = \emptyset$. Then $\{E_i \setminus E_{i-1}\}_{i=1}^\infty$ will be a sequence of pairwise disjoint subsets of X with the property that $\bigcup_{i=1}^\infty E_i = \bigcup_{i=1}^\infty E_i \setminus E_{i-1}$. With this:

$$\begin{aligned}\mu\left(\bigcup_{i=1}^\infty E_i\right) &= \mu\left(\bigcup_{i=1}^\infty E_i \setminus E_{i-1}\right) \\ &= \sum_{i=1}^\infty \mu(E_i \setminus E_{i-1}).\end{aligned}$$

Note that $E_i = E_{i-1} \sqcup (E_i \setminus E_{i-1})$. This means $\mu(E_i) = \mu(E_{i-1}) + \mu(E_i \setminus E_{i-1})$, which is equivalent to

$\mu(E_i \setminus E_{i-1}) = \mu(E_i) - \mu(E_{i-1})$. Hence:

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= \sum_{i=1}^{\infty} \mu(E_i \setminus E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \sum_{n=1}^i \mu(E_i \setminus E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \sum_{n=1}^i \mu(E_i) - \mu(E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \mu(E_i). \end{aligned}$$

(4) Define $F_i = E_1 \setminus E_i$. Then $\{F_i\}_{i=1}^{\infty}$ will be a sequence of increasing unions of subsets of X . In particular, we have that $\bigcup_{i=1}^{\infty} F_i = E_1 \setminus \bigcap_{i=1}^{\infty} E_i$. This means:

$$\begin{aligned} E_1 &= \left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i\right) \sqcup \left(\bigcap_{i=1}^{\infty} E_i\right) \\ &= \left(\bigcup_{i=1}^{\infty} F_i\right) \sqcup \left(\bigcap_{i=1}^{\infty} E_i\right). \end{aligned}$$

Hence by (3):

$$\begin{aligned} \mu\left(\bigcap_{i=1}^{\infty} E_i\right) &= \mu(E_1) - \mu\left(\bigcup_{i=1}^{\infty} F_i\right) \\ &= \mu(E_1) - \lim_{i \rightarrow \infty} \mu(F_i). \end{aligned}$$

But $F_i = E_1 \setminus E_i$, so $E_1 = F_i \sqcup E_i$. This gives $\mu(E_1) = \mu(F_i) + \mu(E_i)$, hence:

$$\begin{aligned} \mu\left(\bigcap_{i=1}^{\infty} E_i\right) &= \mu(E_1) - \lim_{i \rightarrow \infty} \mu(F_i) \\ &= \mu(E_1) - \lim_{i \rightarrow \infty} (\mu(E_1) - \mu(E_i)) \\ &= \lim_{i \rightarrow \infty} \mu(E_i). \end{aligned} \quad \square$$

Definition 1.2.2. Let $(X, \mathcal{M}, \mu)$ be a measure space.

- (1) If $E \in \mathcal{M}$ is a measurable set such that $\mu(E) = 0$, then E is called a null-set.
- (2) If a property holds on all of X except on a null-set, then it holds almost everywhere (abbreviated a.e.).
- (3) If $\mathcal{M}$ has the property that every subset of a null-set is measurable, then $\mathcal{M}$ is said to be complete.

Theorem 1.2.1. Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\mathcal{N} := \{E \in \mathcal{M} \mid \mu(E) = 0\}$. Then the set $\overline{\mathcal{M}} := \{E \cup F \mid E \in \mathcal{M}, F \subset \mathcal{N} \text{ for some } N \in \mathcal{N}\}$ is a σ -algebra. Moreover, there exists a unique function $\overline{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$ which extends μ .

Proof. Consider the family $\{E_i \cup F_i\}_{i=1}^\infty \subset \overline{\mathcal{M}}$, where for each i , there exists $N_i \in \mathcal{N}$ such that $F_i \subset N_i$. Then:

$$\bigcup_{i=1}^\infty E_i \cup F_i = \left(\bigcup_{i=1}^\infty E_i \right) \cup \left(\bigcup_{i=1}^\infty F_i \right);$$

Since $\mathcal{M}$ is a σ -algebra, we have $\bigcup_{i=1}^\infty E_i \in \mathcal{M}$. Moreover, $\bigcup_{i=1}^\infty F_i \subset \bigcup_{i=1}^\infty N_i$. By subadditivity, we have $\mu\left(\bigcup_{i=1}^\infty N_i\right) \leq \sum_{i=1}^\infty \mu(N_i) = 0$. Hence $\bigcup_{i=1}^\infty N_i \in \mathcal{N}$. Together, this shows that $\bigcup_{i=1}^\infty E_i \cup F_i \in \overline{\mathcal{M}}$.

Now let $E \cup F \in \overline{\mathcal{M}}$. Then:

$$\begin{aligned} (E \cup F)^c &= \left((E \cup N) \cap (N^c \cup F) \right)^c \\ &= (E \cup N)^c \cup (N^c \cup F)^c \\ &= (E^c \cap N^c) \cup (N \cap F^c) \\ &= (E^c \cap N^c) \cup (N \setminus F). \end{aligned}$$

Since $E^c, N^c \in \mathcal{M}$, and since $\mathcal{M}$ is closed under countable intersections, we have $E^c \cap N^c \in \mathcal{M}$. Since $N \in \mathcal{N}$, surely $N \setminus F \in \mathcal{N}$, whence $(E \cup F)^c \in \overline{\mathcal{M}}$. Thus $\overline{\mathcal{M}}$ is a σ -algebra.

Define $\bar{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$ by $\bar{\mu}(E \cup F) = \mu(E)$. We must first show that this map is well-defined. Suppose $E_1 \cup F_1 = E_2 \cup F_2$, where $F_i \subset N_i \in \mathcal{N}$ for $i = 1, 2$. Then $E_1 \cup F_1 \subset E_2 \cup N_2$, hence $E_1 \subset E_2 \cup N_2$. This means $\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$. But an identical argument gives $E_2 \subset E_1 \cup N_1$, hence $\mu(E_2) \leq \mu(E_1 \cup N_1) \leq \mu(E_1) + \mu(N_1) = \mu(E_1)$. Together, $\mu(E_1) = \mu(E_2)$, allowing us to show that:

$$\begin{aligned} \bar{\mu}(E_1 \cup F_1) &= \mu(E_1) \\ &= \mu(E_2) \\ &= \bar{\mu}(E_2 \cup F_2). \end{aligned}$$

Thus $\bar{\mu}$ is well-defined. Note that $\emptyset \in \mathcal{M}$ and $\emptyset \in \mathcal{N}$, hence $\bar{\mu}(\emptyset) = \mu(\emptyset) = 0$. Now if $\{E_i \cup F_i\}_{i=1}^\infty \subset \overline{\mathcal{M}}$ are pairwise disjoint, then it must be the case that $\{E_i\}_{i=1}^\infty$ are pairwise disjoint. This gives:

$$\bar{\mu}\left(\bigcup_{i=1}^\infty E_i \cup F_i\right) = \bar{\mu}\left(\bigcup_{i=1}^\infty E_i \cup \bigcup_{i=1}^\infty F_i\right).$$

Note that $\bigcup_{i=1}^\infty E_i \in \mathcal{M}$, and $\bigcup_{i=1}^\infty F_i \subset \bigcup_{i=1}^\infty N_i \in \mathcal{N}$, hence:

$$\begin{aligned} \bar{\mu}\left(\bigcup_{i=1}^\infty E_i \cup F_i\right) &= \bar{\mu}\left(\bigcup_{i=1}^\infty E_i \cup \bigcup_{i=1}^\infty F_i\right) \\ &= \mu\left(\bigcup_{i=1}^\infty E_i\right) \\ &\leq \sum_{i=1}^\infty \mu(E_i) \\ &= \sum_{i=1}^\infty \bar{\mu}(E_i \cup F_i). \end{aligned}$$

Thus $\bar{\mu}$ is a measure. Moreover, it is an extension of μ since for any $E \in \mathcal{M}$, $\bar{\mu}(E) = \bar{\mu}(E \cup \emptyset) = \mu(E)$. It remains to show that this extension is unique. Suppose γ is also an extension of μ onto $\bar{\mathcal{M}}$. (WTS $\gamma(\bar{\mathcal{M}}) = \bar{\mu}(\bar{\mathcal{M}})$). Let $E \cup F \in \bar{\mathcal{M}}$, where $F \subset N \in \mathcal{N}$. By subadditivity:

$$\begin{aligned} \gamma(E \cup F) &\leq \gamma(E) \cup \gamma(F) \\ &= \gamma(E) \\ &= \mu(E) \quad \text{Since } \gamma|_{\mathcal{M}} = \mu \\ &= \bar{\mu}(E \cup F). \end{aligned}$$

But by monotonicity:

$$\begin{aligned} \bar{\mu}(E \cup F) &= \mu(E) \\ &= \gamma(E) \quad \text{Since } \gamma|_{\mathcal{M}} = \mu \\ &\leq \gamma(E \cup F). \end{aligned}$$

Hence $\gamma(E \cup F) = \bar{\mu}(E \cup F)$. Since γ and $\bar{\mu}$ are equal on all values of $\bar{\mathcal{M}}$, they are equal. Thus $\bar{\mu}$ is unique. $\square$

Definition 1.2.3. Let $(X, \mathcal{M}, \mu)$ be a measure space. The measure $\bar{\mu}$ is called the completion of μ , and $\bar{\mathcal{M}}$ is called the completion of $\mathcal{M}$ with respect to μ .

§ 1.3. Outer Measures

Definition 1.3.1. Let X be a nonempty set. An outer measure on X is a function:

$$\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$$

satisfying:

- (1) $\mu^*(\emptyset) = 0$;
- (2) If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$ (Monotonicity);
- (3) $\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (Subadditivity).

Proposition 1.3.1. Let X be any nonempty set and $\mathcal{E} \subset \mathcal{P}(X)$, with $\emptyset, X \in \mathcal{E}$. Let $\rho : \mathcal{E} \rightarrow [0, \infty]$ be a function such that $\rho(\emptyset) = 0$. The function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ defined by:

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset A, E_i \in \mathcal{E} \right\}$$

is an outer measure on X .

Proof. Since $\emptyset \subset \emptyset$ and $\emptyset \in \mathcal{E}$, we get $\mu^*(\emptyset) = 0$. If $A \subset B$ and $\bigcup_{i=1}^{\infty} E_i \supset B$, then surely $\bigcup_{i=1}^{\infty} E_i \supset A$. This gives the containment:

$$\left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset B, E_i \in \mathcal{E} \right\} \supset \left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset A, E_i \in \mathcal{E} \right\}.$$

Any lowerbound for the left set will be a lowerbound for the right set, hence $\mu^*(A) \leq \mu^*(B)$. It remains to show that μ^* is subadditive. Let $\{A_j\}_{j=1}^\infty \subset \mathcal{P}(X)$. Let $\epsilon > 0$. For each $j = 1, 2, 3, \dots$ find a family $\{E_{i,j}\}_{i=1}^\infty \subset \mathcal{E}$ such that:

$$\sum_{i=1}^\infty \rho(E_{i,j}) \leq \frac{\epsilon}{2^j} + \mu^*(A_j). \quad \text{Property of infimums.}$$

Note that $A_j \subset \bigcup_{i=1}^\infty E_{i,j}$, hence $\{E_{i,j}\}_{i,j \in \mathbb{N}}$ covers $\bigcup_{j=1}^\infty A_j$. This means:

$$\begin{aligned} \mu^*\left(\bigcup_{j=1}^\infty A_j\right) &\leq \sum_{j=1}^\infty \sum_{i=1}^\infty \rho(E_{i,j}) \quad \text{Surely any cover will be bigger than the g.l.b.} \\ &\leq \sum_{j=1}^\infty \left(\mu^*(A_j) + \frac{\epsilon}{2^j}\right) \quad \text{Also, rearrangements of sums follows from being non-negative} \\ &= \sum_{j=1}^\infty \mu^*(A_j) + \epsilon. \end{aligned}$$

Taking $\epsilon \rightarrow \infty$ gives $\mu^*\left(\bigcup_{j=1}^\infty A_j\right) \leq \sum_{j=1}^\infty \mu^*(A_j)$. □

This next definition will lead us to constructing measures from outer measures.

Definition 1.3.2. Let μ^* be an outer measure on X . A set $A \subset X$ is called μ^* -measurable if, for all $E \subset X$:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

This definition is very unintuitive. From our example in $\mathbb{R}^2$, given any rectangle R covering $A \subset \mathbb{R}^2$, we defined the inner area of A as:

$$\text{innerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A),$$

where R is a rectangle covering our set A . If the inner area and the outer area are equal, then the above equation becomes:

$$\text{outerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A).$$

Rearranging yields:

$$\text{area}(R) = \text{outerarea}(A) + \text{outerarea}(R \setminus A).$$

So under the assumption that the inner and outer area of A is equal, we gain the ability to measure any rectangle R using *only* the parts which intersect A and the parts which intersect A^c . In this case A is a "nice set" that we can use to measure other things.

Lemma 1.3.1 (Caratheodory's Lemma). *** Let μ^* be an outer measure on X . Let $\mathcal{M} = \{A \subset X \mid A \text{ is } \mu^* \text{ measurable}\}$. Then $\mathcal{M}$ is a σ -algebra, and $\mu^*|_{\mathcal{M}}$ is a complete measure (i.e., $(X, \mathcal{M}, \mu^*|_{\mathcal{M}})$ is a complete measure space).

Proof. 10/2/25 □

Carathéodory's lemma is an extremely powerful result. We will use it to show that a measure defined on an *algebra* can be extended to a measure on a σ -*algebra*. Typically it's easier to construct algebras than σ -algebras, and a direct application of this will be seen in the next section.

Definition 1.3.3. Let X be a nonempty set. Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra of sets. A function $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is a *premeasure* if:

- (1) $\mu_0(\emptyset) = 0$;
- (2) If $\{E_i\}_{i=1}^{\infty} \subset \mathcal{A}$ is a family of pairwise disjoint sets with $\bigcup_{i=1}^{\infty} E_i \in \mathcal{A}$, then $\mu_0\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu_0(E_i)$.

Proposition 1.3.2. If μ_0 is a premeasure on $\mathcal{A}$ and μ^* is the outer measure induced by μ_0 , then $\mu^*|_{\mathcal{A}} = \mu_0$.

Proof. Let $E \in \mathcal{A}$... **finish some day...** □

Theorem 1.3.1 (Carathéodory's Extension Theorem). *** Let μ_0 be a premeasure on an algebra $\mathcal{A} \subset \mathcal{P}(X)$. Then:

- (1) μ_0 can be extended to a measure $\mu : \mathfrak{M}(\mathcal{A}) \rightarrow [0, \infty]$.
- (2) If ν is another measure that extends μ_0 , then $\nu(E) \leq \mu(E)$ for all $E \in \mathfrak{M}(\mathcal{A})$, with equality when $\mu(E) < \infty$.
- (3) If μ_0 is σ -finite, then μ is the unique extension of μ_0 to a measure on $\mathfrak{M}(\mathcal{A})$.

Proof. (1) Our first step will be to build an outer measure on $\mathcal{P}(X)$. We will use the sets from $\mathcal{A}$ as our "measuring sets". For $E \subset X$, define:

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(A_j) \mid E \subset \bigcup_{j=1}^{\infty} A_j, \{A_j\}_{j=1}^{\infty} \subset \mathcal{A} \right\}$$

Note that we can trivially cover E with $X \in \mathcal{A}$, hence it is nonempty. By Proposition 1.3.1, the function μ^* is an outer measure. By Carathéodory's Lemma, we have a measure $\mu^*|_{\mathfrak{M}^*}$ defined on:

$$\mathfrak{M}^* = \{A \subset X \mid A \text{ is } \mu^*\text{-measurable}\}$$

Now that we've constructed $(X, \mathfrak{M}^*, \mu^*|_{\mathfrak{M}^*})$, our next step is to check that $\mathcal{A} \subset \mathfrak{M}^*$ —that is, for any $A \in \mathcal{A}$ and any $E \subset X$, we can write:

$$\mu^*(E) = \mu(E \cap A) + \mu(E \cap A^c).$$

Let $\epsilon > 0$. There exists $\{A_j\}_{j=1}^{\infty} \subset \mathcal{A}$ such that:

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(A_j) && \text{Infimum property} \\ &= \sum_{j=1}^{\infty} \mu_0(A_j \cap (A \cup A^c)) \\ &= \sum_{j=1}^{\infty} (\mu_0(A_j \cap A) + \mu_0(A_j \cap A^c)) \\ &= \sum_{j=1}^{\infty} \mu_0(A_j \cap A) + \sum_{j=1}^{\infty} \mu_0(A_j \cap A^c). \end{aligned}$$

Note that $\{A_j \cap A\}_{j=1}^\infty \subset \mathcal{A}$ covers $E \cap A$ and $\{A_j \cap A^c\}_{j=1}^\infty \subset \mathcal{A}$ covers $E \cap A^c$. Hence:

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{j=1}^\infty \mu_0(A_j \cap A) + \sum_{j=1}^\infty \mu_0(A_j \cap A^c) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \text{Any covering will be larger than g.l.b.} \\ &\geq \mu^*(E). \quad \text{Monotonicity} \end{aligned}$$

Sending $\epsilon \rightarrow 0$ gives equality, hence A is μ^* -measurable. Having established the inclusion $\mathcal{A} \subset \mathcal{M}^*$, we obtain $\mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$.

In summary, we constructed $(X, \mathcal{M}^*, \mu^*|_{\mathcal{M}^*})$ with respect to $\mathcal{A}$, and then showed that $\mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$. Claim: The restriction of μ^* to $\mathfrak{M}(\mathcal{A})$ is an extension of μ_0 , that is, for every $A \in \mathcal{A}$ we have $\mu^*|_{\mathfrak{M}(\mathcal{A})}(A) = \mu_0(A)$. For clarity, define $\mu := \mu^*|_{\mathfrak{M}(\mathcal{A})}$.

Let $A \in \mathcal{A}$. Let $\{\mathcal{O}_j\}_{j=1}^\infty$ be a family of sets defined as follows: $\mathcal{O}_1 = A$ and $\mathcal{O}_j = \emptyset$ for all $j \neq 1$. Then $\{\mathcal{O}_j\}_{j=1}^\infty \subset \mathcal{A}$ is a covering of A , hence $\mu(A) \leq \sum_{j=1}^\infty \mu_0(\mathcal{O}_j) = \mu_0(A)$ (again, because $\mu(A)$ is defined as the infimum of $\sum \mu_0(A_j)$ over all covers $\{A_j\}$ of A , every cover will be greater than or equal to $\mu(A)$). It only remains to show the other direction of the inequality. Let $\epsilon > 0$. Choose a cover $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ of A . Then by properties of infimums once again, we have:

$$\mu(A) + \epsilon \geq \sum_{j=1}^\infty \mu_0(A_j).$$

For each j , define $B_j := A_j \setminus \bigcup_{k=1}^{j-1} A_k \in \mathcal{A}$. This construction was seen in Proposition ??, where we showed that $\{B_j\}_{j=1}^\infty$ is a disjoint family of subsets, $B_j \subset A_j$ for each j , and $\bigcup_{j=1}^\infty B_j = \bigcup_{j=1}^\infty A_j$. Observe that:

$$\begin{aligned} \mu(A) + \epsilon &\geq \sum_{j=1}^\infty \mu_0(A_j) \\ &\geq \sum_{j=1}^\infty \mu_0(B_j) \quad \text{Monotonicity of premeasures} \\ &\geq \sum_{j=1}^\infty \mu_0(A \cap B_j) \quad \text{Monotonicity of premeasures} \\ &= \mu_0\left(\bigcup_{j=1}^\infty A \cap B_j\right) \quad \text{Premeasures are countably additive over disjoint sets} \\ &= \mu_0\left(A \cap \bigcup_{j=1}^\infty B_j\right) \quad \text{DeMorgan's Law} \\ &= \mu_0\left(A \cap \bigcup_{j=1}^\infty A_j\right) \quad \text{By Proposition ??} \\ &= \mu_0(A). \quad \text{Since } \{A_j\}_{j=1}^\infty \text{ is a cover of } A \end{aligned}$$

Taking $\epsilon \rightarrow \infty$ gives $\mu(A) \geq \mu_0(A)$. Thus $\mu(A) = \mu_0(A)$.

(b) Let $E \in \mathfrak{M}(\mathcal{A})$ and $\epsilon > 0$. Choose a cover $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ of E such that:

$$\begin{aligned} \mu(E) + \epsilon &\geq \sum_{j=1}^\infty \mu_0(A_j) \\ &= \sum_{j=1}^\infty \mu(A_j). \quad \text{Since } \mu|_{\mathcal{A}} = \mu_0 \end{aligned}$$

Then:

$$\begin{aligned} \nu(E) &\leq \nu\left(\bigcup_{j=1}^\infty A_j\right) \\ &\leq \sum_{j=1}^\infty \nu(A_j) \\ &= \sum_{j=1}^\infty \mu_0(A_j) \\ &\leq \mu(E) + \epsilon. \end{aligned}$$

Taking $\epsilon \rightarrow 0$ gives $\nu(E) \leq \mu(E)$. Now suppose $\mu(E) < \infty$. Let $\epsilon > 0$. Find a cover $A := \bigcup_{j=1}^\infty A_j \supset E$ such that $\mu(E) + \epsilon \geq \sum_{j=1}^\infty \mu_0(A_j) \geq \mu(A)$. Rearranging gives $\mu(A) - \mu(E) \leq \epsilon$, but since $\mu(E) < \infty$ we can write this as $\mu(A \setminus E) \leq \epsilon$. **Finish the rest soon** $\square$

§ 1.4. Measures on $\mathbb{R}$

Carathéodory's Extension Theorem will finally allow us to construct measures on the real number line. As we showed in §2.0, we are unable to define measures on the power set of $\mathbb{R}$. Our game-plan is to find an algebra $\mathcal{A}$ of subsets of $\mathbb{R}$ such that, after applying Carathéodory's Extension Theorem, we obtain a measure on the Borel sets of $\mathbb{R}$.

Definition 1.4.1. Let X be a set. An elementary family is a collection $\mathcal{E}$ of subsets of X such that:

- (1) $\emptyset \in \mathcal{E}$;
- (2) if $E, F \in \mathcal{E}$, then $E \cap F \in \mathcal{E}$;
- (3) if $E \in \mathcal{E}$, then E^c is a finite disjoint union of members of $\mathcal{E}$.

Proposition 1.4.1. If $\mathcal{E}$ is an elementary family, the collection $\mathcal{A}$ of finite disjoint unions of members of $\mathcal{E}$ is an algebra.

The proof of this proposition is not relevant to our current discussion.

Definition 1.4.2. An h -interval is a left-open, right-closed interval in $\mathbb{R}$ —that is, sets of the form $(a, b]$, (a, ∞) or $\emptyset$.

These h -intervals will be the building blocks for constructing a measure on $\mathbb{R}$. Clearly the intersection of two h -intervals is an h -interval, and the complement of an h -interval is an h -interval or the union of two h -intervals. Moreover, by Proposition 1.4.1, the collection of *finite disjoint unions of h -intervals* is an algebra. This collection can be used to generate $\mathcal{B}_{\mathbb{R}}$, as per Proposition 1.1.4

Proposition 1.4.2. *Let μ be a finite measure on $\mathcal{B}_{\mathbb{R}}$. Define $F : \mathbb{R} \rightarrow \mathbb{R}$ by $x \mapsto \mu((-\infty, x])$. Then F is increasing and right-continuous. Moreover, if $a < b$, we have $\mu((a, b]) = F(b) - F(a)$.*

Proof. F increasing follows from monotonicity of measures; if $x \leq y$ are real numbers, then it must be that $F(x) = \mu((-\infty, x]) \leq \mu((-\infty, y]) = F(y)$. If $(x_n)_n$ is a sequence in $\mathbb{R}$ such that $(x_n)_n \searrow x$, then:

$$\begin{aligned} F(x) &= \mu((-\infty, x]) \\ &= \mu\left(\bigcap_{n=1}^{\infty} (-\infty, x_n]\right) \\ &= \lim_{n \rightarrow \infty} \mu((-\infty, x_n]) \quad \text{Continuity from above} \\ &= \lim_{n \rightarrow \infty} F(x_n). \end{aligned}$$

Hence F is right-continuous. Finally, if $a < b$, we can write $(-\infty, b] = (-\infty, a] \cup (a, b]$. Since μ is a finite measure, by countable additivity we have $\mu((-\infty, b]) = \mu((-\infty, a]) + \mu((a, b])$. Rearranging gives $\mu((a, b]) = \mu((-\infty, b]) - \mu((-\infty, a]) = F(b) - F(a)$. $\square$

Definition 1.4.3. *The function F described in Proposition 1.4.2 is called a distribution function of μ .*

The distribution function of a measure provides information about the size of a measurable function.

Proposition 1.4.3. *Let $\mathcal{A}$ be the collection of finite disjoint unions of h -intervals. Let F be a distribution function of $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$, where μ_0 satisfies the following:*

(1) $\mu_0(\emptyset) = 0$;

(2) *If $\{(a_j, b_j]\}_{j=1}^n$ are disjoint h -intervals, then $\mu_0\left(\bigcup_{j=1}^n (a_j, b_j]\right) = \sum_{j=1}^n (F(b_j) - F(a_j))$.*

Then μ_0 is a premeasure on the algebra $\mathcal{A}$.

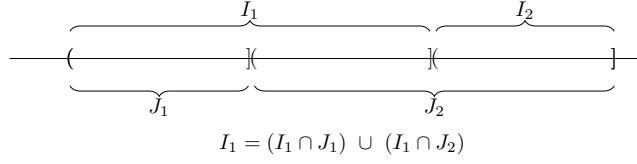
Proof. We must first show that μ_0 is well-defined. We will first prove a special case of μ_0 being well-defined, and then move onto a more general case. Suppose $(a, b] = \bigcup_{j=1}^m (a_j, b_j]$, where each $\{(a_j, b_j]\}_{j=1}^m$ are disjoint. Without loss of generality, assume $a_1 < a_2 < \dots < a_m$. It must be the case that $b_1 = a_2$, $b_2 = a_3$, etc. With this, note that the sum $\sum_{j=1}^m (F(b_j) - F(a_j))$ is telescoping, hence $\mu_0\left(\bigcup_{j=1}^m (a_j, b_j]\right) = \sum_{j=1}^m (F(b_j) - F(a_j)) = F(b) - F(a) = \mu_0((a, b])$. Thus, under this special case, μ_0 is well-defined. For the general case, suppose $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^n$ are collections of disjoint h -intervals such that $\bigcup_{i=1}^n I_i = \bigcup_{j=1}^n J_j$. Note that the intersections $I_i \cap J_j$ are (possibly empty) h -intervals, and they form a disjoint partition of each I_i and each J_j :

$$I_i = \bigcup_{j=1}^n (I_i \cap J_j), \quad J_j = \bigcup_{i=1}^n (I_i \cap J_j).$$

(See Figure 2.1 as a motivating example). Apply the special case inside each I_i and J_j :

$$\mu_0(I_i) = \sum_{j=1}^n \mu_0(I_i \cap J_j), \quad \mu_0(J_j) = \sum_{i=1}^n \mu_0(I_i \cap J_j).$$

Summing each equation and rearranging gives $\sum_{i=1}^n \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_{j=1}^n \mu_0(J_j)$. Thus μ_0 is well-defined.

Figure 1.1: Illustration of interval intersections between I_1 , I_2 , and J_1 , J_2 .

We only need to show that μ_0 satisfies countable additivity. Let $\{A_j\}_{j=1}^{\infty} \subset \mathcal{A}$ be a family of disjoint sets, where $\bigcup_{j=1}^{\infty} A_j \in \mathcal{A}$. Since everything in $\mathcal{A}$ is a finite disjoint union of h-intervals, we can write:

$$\bigcup_{j=1}^{\infty} A_j = \bigcup_{n=1}^N I_n, \quad A_j = \bigcup_{i=1}^{N_j} I_{i,j}.$$

But observe that:

$$\begin{aligned} I_n &= I_n \cap \left(\bigcup_{j=1}^{\infty} A_j \right) \\ &= \bigcup_{j=1}^{\infty} (I_n \cap A_j) \\ &= \bigcup_{j=1}^{\infty} \left(I_n \cap \left(\bigcup_{i=1}^{N_j} I_{i,j} \right) \right) \\ &= \bigcup_{j=1}^{\infty} \bigcup_{i=1}^{N_j} (I_n \cap I_{i,j}). \end{aligned}$$

It is enough to assume that if an h-interval I is a countable disjoint union of h-intervals $\{I_j\}_{j=1}^{\infty}$, then $\mu_0(I) = \sum_{j=1}^{\infty} \mu_0(I_j)$. Let $I = (a, b]$ and $I_j = (a_j, b_j]$. One direction of the equality can be seen as follows:

$$\begin{aligned} \mu_0(I) &= \mu_0 \left(\bigcup_{j=1}^n I_j \right) + \mu_0 \left(I \setminus \bigcup_{j=1}^n I_j \right) \\ &\geq \mu_0 \left(\bigcup_{j=1}^n I_j \right) \\ &= \sum_{j=1}^n \mu_0(I_j). \end{aligned}$$

Taking $n \rightarrow \infty$ gives $\mu_0(I) \geq \sum_{j=1}^{\infty} \mu_0(I_j)$. Conversely, let $\epsilon > 0$. By the right-continuity of F , choose $\delta > 0$ such that $F(a + \delta) - F(a) < \epsilon$. Similarly, choose $\delta_j > 0$ such that $F(b_j + \delta_j) - F(b_j) < \frac{\epsilon}{2^j}$. Then the compact set $[a + \delta, b]$ is covered by the open intervals $\{(a_j, b_j + \delta_j)\}_{j=1}^{\infty}$, so there exists a finite subcover $\{(a_j, b_j)\}_{j=1}^N$. Without loss of generality, let $a_1 < a_2 < \dots < a_N$. By construction

$b_j + \delta_j \in (a_{j+1}, b_{j+1} + \delta_{j+1})$. We then get:

$$\begin{aligned}
 \mu_0(I) &= F(b) - F(a) \\
 &\leq F(b) - F(a + \delta) + \epsilon \\
 &= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(a_{j+1}) - F(a_j)) + \epsilon \\
 &\leq F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(b_j + \delta_j) - F(a_j)) + \epsilon \\
 &= \sum_{j=1}^N (F(b_j + \delta_j) - F(a_j)) + \epsilon \\
 &< \sum_{j=1}^N (F(b_j) + \frac{\epsilon}{2^j} - F(a_j)) + \epsilon \\
 &\leq \sum_{j=1}^N (F(b_j) - F(a_j)) + 2\epsilon \\
 &= \sum_{j=1}^N \mu_0(I_j) + 2\epsilon
 \end{aligned}$$

Taking $\epsilon \rightarrow 0$ finally shows that μ_0 is countable additive over disjoint unions of h -intervals. Thus, μ_0 is a well-defined premeasure defined on $\mathcal{A}$, the algebra of countable disjoint unions of h -intervals. $\square$

Having finally constructed the premeasure $(\mathbb{R}, \mathcal{A}, \mu_0)$, we can now apply Carathéodory's Extension Theorem.

Theorem 1.4.1. *Let F be a distribution function. There is a unique Borel measure μ_F on $\mathbb{R}$ such that $\mu_F((a, b]) = F(b) - F(a)$. If G is another such function, then $\mu_F = \mu_G$ if and only if $F - G$ is constant. Conversely, if μ is a Borel measure on $\mathbb{R}$ that is finite on all bounded Borel sets, define:*

$$F(x) = \begin{cases} \mu((0, x]), & x > 0, \\ 0, & x = 0, \\ \mu(-(x, 0]), & x < 0, \end{cases}$$

Then F is a distribution, and $\mu = \mu_F$.

Proof. Proposition 1.4.3 showed that each F induces a premeasure on $\mathcal{A}$, the collection of countable disjoint unions of h -intervals. It is clear that F and G induce the same premeasure if $F - G$ is constant, and that these premeasures are σ -finite. The assertion that there is a *unique* Borel measure comes from Carathéodory's Extension Theorem. Given the outer measure μ^* , induced by the premeasure on countable disjoint unions of h -intervals, define $\mu_F := \mu^*|_{\mathfrak{M}(\mathcal{A})} = \mu^*|_{\mathcal{B}_{\mathbb{R}}}$. The converse direction follows from the monotonicity of μ . $\square$

There are many levels to this application of Theorem 1.3.1, Theorem 1.3.1, and Proposition 1.4.3. Different restrictions of the outer-measure space $(X, \mathcal{P}(X), \mu^*)$ give different σ -algebra to work in.

We have the chain of σ -algebras:

$$\mathcal{A} \subset \mathcal{B}_{\mathbb{R}} \subsetneq \mathcal{M}^* \subset \mathcal{P}(X),$$

Where $\mu_F := \mu^*|_{\mathcal{M}(\mathcal{A})} = \mu^*|_{\mathcal{B}_{\mathbb{R}}}$ is the measure on $\mathcal{B}_{\mathbb{R}}$, and $\mu^*|_{\mathcal{M}^*}$ is the *complete* measure on $\mathcal{M}^*$, the set of all μ^* -measurable sets. One can rigorously show that the domain of $\mu^*|_{\mathcal{M}^*}$ is strictly larger than the domain of $\mu^*|_{\mathcal{B}_{\mathbb{R}}}$.

Definition 1.4.4. Let μ_0 be a premeasure on $\mathcal{A}$ (the algebra of countable disjoint unions of h -intervals) and F a distribution function on μ_0 . The restriction of the $(X, \mathcal{A}, \mu_0)$ induced outer measure onto all measurable sets is called the Lebesgue-Stieltjes measure associated to F .

For the remainder of this section, let μ denote the Lebesgue-Stieltjes measure on $\mathbb{R}$ associated to the distribution function F , $\mathcal{M}_{\mu}$ the domain of μ , and $\mathcal{A}$ the algebra of countable disjoint unions of h -intervals. We can rewrite our standard definition of the outer measure as follows:

$$\begin{aligned} \mu(E) &= \inf \left\{ \sum_{j=1}^{\infty} \mu_0(A_j) \mid \{A_j\}_{j=1}^{\infty} \subset \mathcal{A}, \bigcup_{j=1}^{\infty} A_j \supset E \right\} \\ &= \inf \left\{ \sum_{j=1}^{\infty} \mu((a_j, b_j]) \mid \{(a_j, b_j]\}_{j=1}^{\infty} \subset \mathcal{A}, \bigcup_{j=1}^{\infty} (a_j, b_j] \supset E \right\}. \end{aligned}$$

Note that $\mu|_{\mathcal{A}} = \mu_0$, hence the second equality.

Theorem 1.4.2. If $E \in \mathcal{M}_{\mu}$, then:

- (1) $\mu(E) = \inf\{\mu(U) \mid U \supset E \text{ and } U \text{ is open}\}.$
- (2) $\mu(E) = \sup\{\mu(K) \mid K \subset E \text{ and } K \text{ is compact}\}.$

Proof. (1) Note that $\mu(E) \leq \inf\{\mu(U) \mid U \supset E \text{ and } U \text{ is open}\}$ follows by monotonicity since $E \subset U$. For the other direction, let $\epsilon > 0$. Choose h -intervals $\{I_j\}_{j=1}^{\infty} := \{(a_j, b_j]\}_{j=1}^{\infty} \subset \mathcal{A}$ such that $\bigcup_{j=1}^{\infty} I_j \supset E$ and:

$$\mu(E) + \epsilon \geq \sum_{j=1}^{\infty} \mu(I_j).$$

By the right continuity of F (our underlying distribution function), for each j choose $\delta_j > 0$ such

that $F(b_j + \delta_j) - F(b_j) < \frac{\epsilon}{2^j}$. Observe that:

$$\begin{aligned}
 \mu(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu(I_j) \\
 &= \sum_{j=1}^{\infty} \mu((a_j, b_j + \delta_j) \setminus (b_j, b_j + \delta_j)) \\
 &= \sum_{j=1}^{\infty} (\mu((a_j, b_j + \delta_j)) - \mu((b_j, b_j + \delta_j))) \\
 &\geq \sum_{j=1}^{\infty} (\mu((a_j, b_j + \delta_j)) - \mu([b_j, b_j + \delta_j])) \quad \text{Threw in endpoint, hence the inequality} \\
 &\geq \sum_{j=1}^{\infty} (\mu((a_j, b_j + \delta_j)) - (F(b_j + \delta_j) - F(b_j))) \\
 &\geq \sum_{j=1}^{\infty} (\mu((a_j, b_j + \delta_j)) - \frac{\epsilon}{2^j}) \\
 &= \sum_{j=1}^{\infty} (\mu((a_j, b_j + \delta_j))) - \epsilon \\
 &\geq \mu\left(\bigcup_{j=1}^{\infty} (a_j, b_j + \delta_j)\right) - \epsilon
 \end{aligned}$$

Take $U_\epsilon := \bigcup_{j=1}^{\infty} (a_j, b_j + \delta_j)$. Then $\mu(E) \geq \mu(U_\epsilon) + \epsilon$. Taking $\epsilon \rightarrow \infty$ gives the desired result.

(2) We proceed by cases on E . Case 1: E is bounded. Apply part (1) to the set $\bar{E} \setminus E$. Let $\epsilon > 0$. Find an open set $U \supset \bar{E} \setminus E$ such that $\mu(U) - \mu(\bar{E} \setminus E) \leq \epsilon$. Define $K = \bar{E} \setminus U$. Note that K is bounded since $\bar{E} \setminus U = \bar{E} \cap U^c$ is a closed subset of $\bar{E}$, which is a compact set. We can write:

$$\begin{aligned}
 K &= \bar{E} \setminus U \\
 &= (E \cup (\bar{E} \setminus E)) \setminus U \\
 &= E \setminus U \\
 &\subset E.
 \end{aligned}$$

Note that $E \setminus K \subset U \setminus (\bar{E} \setminus E)$ * (the dotted line apparently). So $\mu(E \setminus K) \leq \mu(U \setminus (\bar{E} \setminus E))$, which is equivalent to $\mu(E) - \mu(K) \leq \mu(E) - \mu(\bar{E} \setminus E) \leq \epsilon$. Rearranging and taking $\epsilon \rightarrow 0$ gives $\mu(E) \leq \mu(K)$, which implies $\mu(E) \leq \sup\{\mu(K) \mid K \subset E \text{ and } K \text{ is compact}\}$. The reverse direction is obvious by monotonicity.

Case 2: E is unbounded. Again, one direction is obvious by monotonicity. Let $E_j = E \cap (j, j+1]$. Then $\mu(E) = \sum_{j=1}^{\infty} \mu(E_j)$. For all j , let $K_j \subset E_j$ be compact, and:

$$\mu(E) - \mu(K_j) < \frac{\epsilon}{2^{|j|}}.$$

Then:

$$\begin{aligned}
 \mu(E_j) &= \sum_{j \in \mathbb{Z}} \mu(E_j) \\
 &\leq \sum_{j=-\infty}^{\infty} \left(\mu(K_j) + \frac{\epsilon}{2^{|j|}} \right) \\
 &= 3\epsilon + \sum_{j=-\infty}^{\infty} \mu(K_j) \\
 &= 3\epsilon + \lim_{n \rightarrow \infty} \mu \left(\bigcup_{i=-n}^n K_i \right)
 \end{aligned}$$

Note that finite unions of compact sets is compact. Taking $\epsilon \rightarrow 0$ gives the desired result. $\square$

Theorem 1.4.3. *Let $E \subset \mathbb{R}$ be of finite measure. The following are equivalent:*

- (1) $E \in \mathcal{M}_\mu$;
- (2) $E = V \setminus N_1$, where V is a G_δ set and $\mu(N_1) = 0$.
- (3) $E = H \cup N_2$, where H is an F_σ set and $\mu(N_2) = 0$.

Proof. (1) $\Rightarrow$ (2) Let $E \in \mathcal{M}_\mu$. By Theorem 1.4.2, there exists open sets $\{U_n\}_{n \in \mathbb{N}}$ such that:

$$\mu(U_n) < \mu(E) + \frac{1}{2^n}.$$

Since $\mu(E) < \infty$, we can write this as $\mu(U_n) - \mu(E) < \frac{1}{2^n}$. Define $V = \bigcap_{n \in \mathbb{N}} U_n$. Then V is a G_δ set. Since $E \subset U_n$ for all $n \in \mathbb{N}$, we have $E \subset V$. By monotonicity, we have $\mu(E) \leq \mu(V)$. But we also have:

$$\mu(V) \leq \mu(U_n) \leq \mu(E) + \frac{1}{2^n}.$$

Taking $\epsilon \rightarrow 0$ gives $\mu(V) \leq \mu(E)$. Together, $\mu(V) = \mu(E)$; i.e., $\mu(V \setminus E) = 0$. Define $N_1 = V \setminus E$. Then $E = V \setminus N_1$, where V is a G_δ set and N_1 is a null-set.

(1) $\Rightarrow$ (3). Let $E \in \mathcal{M}_\mu$. By Theorem 1.4.2, there exists compact sets $\{K_n\}_{n \in \mathbb{N}}$ such that:

$$\mu(E) - \frac{1}{2^n} < \mu(K_n).$$

Define $H = \bigcup_{n \in \mathbb{N}} K_n$. Then H is a F_σ set. Since $K_n \subset E$ for all $n \in \mathbb{N}$, we have $H \subset E$. By monotonicity, we have $\mu(H) \leq \mu(E)$. But we also have:

$$\mu(E) - \frac{1}{2^n} < \mu(K_n) \leq \mu(H).$$

Taking $n \rightarrow \infty$ gives $\mu(E) \leq \mu(H)$. Together, $\mu(E) = \mu(H)$; i.e., $\mu(E \setminus H) = 0$. Define $N_2 = E \setminus H$. Then $E = H \cup N_2$, where H is an F_σ set and N_2 is a null-set.

(2) $\Rightarrow$ (1) Suppose $E = V \setminus N_1$, where V is a G_δ set and N_1 is a null-set. Since V is a G_δ set, $V \in \mathcal{M}_\mu$. Since N_1 is a null-set, it is measurable. Since $\mathcal{M}_\mu$ is a

(3) $\Rightarrow$ (1) This follows similarly as above. $\square$

Since the countable union of G_δ sets may not be a G_δ set, it is not straight forward to use the fact μ is σ -finite to extend the result for sets of finite measures to the general case. In fact, proving the general case is more like re-doing the proof completely than simply extending the result for sets of finite measures.

Proposition 1.4.4. *If $E \in \mathcal{M}_\mu$ and $\mu(E) < \infty$, then for every $\epsilon > 0$ there is a set A that is a finite union of open intervals such that $\mu(E \Delta A) < \epsilon$.*

Proof. Since $E \in \mathcal{M}_\mu$, $\mu(E) = \inf\{\mu(U) \mid U \supset E \text{ and } U \text{ is open}\}$. Let $\epsilon > 0$. Find an open set U such that $U \supset E$ and:

$$\mu(U) < \mu(E) + \frac{\epsilon}{2}.$$

Since U is open, there exists disjoint open intervals $\{U_j\}_{j=1}^\infty$ such that $\bigcup_{j=1}^\infty U_j = U$. It follows that:

$$\begin{aligned} \mu(U) &= \mu\left(\bigcup_{j=1}^\infty U_j\right) \\ &= \sum_{j=1}^\infty \mu(U_j). \end{aligned}$$

Note that this series converges since $\mu(U) < \mu(E) + \frac{\epsilon}{2}$. Pick N large so that:

$$\sum_{j=1}^{N-1} \mu(U_j) < \frac{\epsilon}{2}.$$

Define $A = \bigcup_{j=1}^{N-1} U_j$. Then $A \subset U$. Observe that:

$$\begin{aligned} \mu(E \Delta A) &= \mu((E \setminus A) \cup (A \setminus E)) \\ &\leq \mu(E \setminus A) + \mu(A \setminus E) \end{aligned}$$

should finish □

We will now start discussing the most important measure on $\mathbb{R}$.

Definition 1.4.5. *The Lebesgue-Stieltjes measure on $\mathbb{R}$ associated to the distribution function $F(x) = x$ is called the Lebesgue measure on $\mathbb{R}$, and is denoted by m . The domain of m is the collection of μ^* -measurable sets, which we will denote as $\mathcal{L}$.*

Sometimes we may also refer to the domain of $\mu^*|_{\mathcal{B}_\mathbb{R}}$ as the Lebesgue measure. This will get confusing at times.

Theorem 1.4.4. *Let $E \in \mathcal{L}$. For all $s, r \in \mathbb{R}$, we have $E + s \in \mathcal{L}$ and $rE \in \mathcal{L}$. Moreover, $m(E + s) = m(E)$ and $m(rE) = |r|m(E)$.*

Proof. Every open set in $\mathbb{R}$ can be expressed as the countable union of disjoint open intervals. ... □

Chapter 2

Integration

From a categorical perspective, now that we have studied measure spaces, we turn our attention to the maps between them.

§ 2.1. Measurable Functions

Definition 2.1.1. Let X be a set and $\mathcal{M}$ a σ -algebra on X . We call the pair $(X, \mathcal{M})$ a measurable space.

Note that a *measure space* has a measure associated to it.

Definition 2.1.2. Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. A function $f : X \rightarrow Y$ is said to be $(\mathcal{M}, \mathcal{N})$ -measurable if for every $E \in \mathcal{N}$, we have $f^{-1}(E) \in \mathcal{M}$.

Proposition 2.1.1.

- (1) If $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable and $g : Y \rightarrow Z$ is $(\mathcal{N}, \mathcal{O})$ -measurable, then $g \circ f : X \rightarrow Z$ is $(\mathcal{M}, \mathcal{O})$ -measurable.
- (2) Let $\mathcal{E}$ be a set and suppose $(X, \mathcal{M})$ and $(Y, \mathfrak{M}(\mathcal{E}))$ are measurable spaces. A function $f : X \rightarrow Y$ is $(\mathcal{M}, \mathfrak{M}(\mathcal{E}))$ -measurable if and only if for every $E \in \mathcal{E}$ we have $f^{-1}(E) \in \mathcal{M}$.
- (3) Let X, Y be metric spaces. If $f : X \rightarrow Y$ is continuous, then f is $(\mathcal{B}_X, \mathcal{B}_Y)$ -measurable.
- (4) Let $(Y_1, \mathcal{N}_1)$ and $(Y_2, \mathcal{N}_2)$ be measurable spaces. The projection maps:

$$Y_1 \times Y_2 \xrightarrow{\pi_i} Y_i, \quad i = 1, 2$$

are $(\mathcal{N}_1 \otimes \mathcal{N}_2, \mathcal{N}_i)$ -measurable.

- (5) The function $f : X \rightarrow Y_1 \times Y_2$ is $(\mathcal{M}, \mathcal{N}_1 \otimes \mathcal{N}_2)$ -measurable if and only if $\pi_1 \circ f$ is $(\mathcal{M}, \mathcal{N}_1)$ -measurable and $\pi_2 \circ f$ is $(\mathcal{M}, \mathcal{N}_2)$ -measurable.

We will be focusing on measurable functions whose codomain is a metric space, hence the following definition.

Definition 2.1.3.

- (1) Let $(X, \mathcal{M})$ be a measurable space and Y a metric space. A function $f : X \rightarrow Y$ is called $\mathcal{M}$ -measurable if it is $(\mathcal{M}, \mathcal{B}_Y)$ -measurable.
- (2) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called Lebesgue measurable if it is $(\mathcal{L}, \mathcal{B}_{\mathbb{R}})$ -measurable.

Remark. Note that the composition of two Lebesgue measurable functions is *not* Lebesgue measurable. If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are both Lebesgue measurable, then their composition $g \circ f$ is $(\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{R}})$ -measurable by Proposition 2.1.1.

Proposition 2.1.2. *Let $(X, \mathcal{M})$ be a measurable space and $f : X \rightarrow \mathbb{C}$ a map. Then f is $\mathcal{M}$ -measurable if and only if $\Re(f)$ and $\Im(f)$ are $\mathcal{M}$ -measurable.*

Proof. Recall from Proposition ?? that $\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} = \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$. Define $\Phi : \mathbb{C} \rightarrow \mathbb{R}^2$ by $a + bi \mapsto (a, b)$. Then $\Phi \circ f = (\Re(f), \Im(f))$. **I don't understand this shit at all** $\square$

Proposition 2.1.3. *If $f, g : X \rightarrow \mathbb{R}$ are both $\mathcal{M}$ -measurable, then $f + g$ and fg are $\mathcal{M}$ -measurable.*

Proof. Consider the composition of maps:

$$X \xrightarrow{(f,g)} \mathbb{R} \times \mathbb{R} \xrightarrow{P} \mathbb{R},$$

where:

$$(f, g)(x) = (f(x), g(x)), \quad P(x, y) = x + y.$$

It is well-known that P is continuous, hence P is $(\mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{R}})$ -measurable. Moreover, (f, g) is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}})$ -measurable because —by part (5) of Proposition 2.1.1 —we have $\pi_1 \circ (f, g) = f$ and $\pi_2 \circ (f, g) = g$. Then by part (1) of Proposition 2.1.1, both f and g are $\mathcal{M}$ -measurable. The argument for fg is identical. $\square$

A similar argument follows for when f and g are complex valued functions, however great care needs to be taken when considering functions whose codomain is the extended reals. It is tedious to avoid $\infty - \infty$.

Proposition 2.1.4. *Let $(X, \mathcal{M})$ be a measurable space. The following are equivalent:*

- (1) $f : X \rightarrow \mathbb{R}$ is $\mathcal{M}$ -measurable;
- (2) For all $x \in \mathbb{R}$, $f^{-1}((x, \infty)) \in \mathcal{M}$;
- (3) For all $x \in \mathbb{R}$, $f^{-1}([x, \infty)) \in \mathcal{M}$;
- (4) For all $x \in \mathbb{R}$, $f^{-1}((-\infty, x)) \in \mathcal{M}$;
- (5) For all $x \in \mathbb{R}$, $f^{-1}((-\infty, x]) \in \mathcal{M}$.

Proof. Come back, something about the homework 1 problem $\square$

Talk about borel sets of extended reals. Then explain that the above proposition but for extended real valued functions follows similarly, since you can show that borel sets of extended reals is generated by the closed endpoint rays.

Proposition 2.1.5. *Let $(X, \mathcal{M})$ be a measurable space and $(f_j)_{j=1}^{\infty}$ be a sequence of functions in $\overline{\mathbb{R}}^X$ which are $\mathcal{M}$ -measurable for each $j \in \mathbb{N}$. The functions:*

$$\sup_j f_j, \quad \inf_j f_j, \quad \limsup_j f_j, \quad \liminf_j f_j$$

are all $\mathcal{M}$ -measurable.

Proof. :(10/15 $\square$

talk about why if its complex valued then its also measurable.

Proposition 2.1.6. *If $f, g : X \rightarrow \overline{\mathbb{R}}$ are $\mathcal{M}$ -measurable, then $\max\{f, g\}$ and $\min\{f, g\}$ are $\mathcal{M}$ -measurable.*

Proof. max and min can be expressed differently or smth like that □

Definition 2.1.4. *If $f : X \rightarrow \overline{\mathbb{R}}$, we define the positive and negative parts to be:*

$$f^+ = \max\{f, 0\}, \quad f^- = \max\{-f, 0\}.$$

Note that $f = f^+ - f^-$, hence if f^+ and f^- are measurable, then f is measurable.

§ 2.2. Integration

Recall that the Riemann Integral involves partitioning the domain of our function into disjoint intervals, and looking at the distance of these step functions and multiplying it by their height.
 explain intuitively what we want the lebesgue integral to do.

For the remainder of this section, fix a measure space $(X, \mathcal{M}, \mu)$.

Definition 2.2.1. *The characteristic function of a subset $E \subset X$ is defined as:*

$$\chi_E(x) = \begin{cases} 1, & x \in E \\ 0, & x \notin E. \end{cases}$$

Proposition 2.2.1. *Let $E \subset X$. Then χ_E is $\mathcal{M}$ -measurable if and only if $E \in \mathcal{M}$.*

Proof. :(10/14 □

(motivate simple functions as a way to talk about integration???)

Definition 2.2.2.

- (1) A simple-function φ is a finite linear combination of $\mathcal{M}$ -measurable characteristic functions.
- (2) The standard representation of a simple function φ is:

$$\varphi = \sum_{j=1}^n z_j \chi_{E_j}, \text{ where } E_j = \varphi^{-1}(\{z_j\}) \text{ and } \text{im } \varphi = \{z_1, \dots, z_n\}.$$

Note that the standard representation exhibits φ as a linear combination of distinct coefficients, as well as the E_j forming a partition of X . It is clear that the set of all simple functions from X to $\mathbb{C}$ is a $\mathbb{C}$ -algebra.

Theorem 2.2.1. *Let $f : X \rightarrow [0, \infty]$ be $\mathcal{M}$ -measurable. There exists a sequence of simple functions $(\varphi_n)_n$ such that $0 \leq \varphi_1 \leq \varphi_2 \leq \dots \leq f$ and $\varphi_n \rightarrow f$. If f is bounded, then this convergence is uniform.*

Proof. :(10/15 □

The above proof is hard to read, but it is easily understood graphically:
 include graphics here :(

Definition 2.2.3.

(1) The space of all measurable functions from X to $[0, \infty]$ is denoted:

$$L^+ := \{f : X \rightarrow [0, \infty] \mid f \text{ is } \mathcal{M}\text{-measurable}\}$$

(2) The subset of L^+ which contains only simple functions is denoted:

$$S^+ := \{\varphi : X \rightarrow [0, \infty] \mid \varphi \in L^+, \varphi \text{ is a simple function}\}.$$

Let $\varphi = \sum_{j=1}^n a_j \chi_{E_j}$ be a simple function in standard representation. Intuitively, the area under this curve should be $\sum_{j=1}^n a_j \mu(E_j)$. Moreover, the area under the curve of any measurable function f should be the "best" approximation of area under a simple function strictly contained between f and 0. **This sounds fucking terrible**

Definition 2.2.4.

(1) For any $\varphi \in S^+$ in standard representation, the integral of φ is:

$$\int \varphi d\mu := \sum_{j=1}^n a_j \mu(E_j).$$

(2) For any $f \in L^+$, the integral of f is:

$$\int f d\mu := \sup \left\{ \int \varphi d\mu \mid 0 \leq \varphi \leq f, \varphi \in S^+ \right\}.$$

(3) If $A \in \mathcal{M}$, the integral of f over A is:

$$\int_A f d\mu := \int f \chi_A d\mu.$$

If $\mathcal{M}$ is a σ -algebra and $E, F \in \mathcal{M}$, it should be noted that $\chi_E \cdot \chi_F = \chi_{E \cap F}$. Hence if φ is a simple function in standard representation and $A \in \mathcal{M}$, we have:

$$\begin{aligned} \varphi \cdot \chi_A &= \left(\sum_{i=1}^n a_i \chi_{E_i} \right) \cdot \chi_A \\ &= \sum_{i=1}^n a_i \chi_{E_i \cap A}. \end{aligned}$$

We will often abuse notation in the following way: if $(X, \mathcal{M}, \mu)$ is a measure space and $f \in L^+$, then:

$$\int f := \int f d\mu := \int_X f d\mu := \int_X f(x) d\mu(x)$$

are all equivalent ways of saying the "integral of f ". We will mostly be taking the minimalistic approach of $\int f$ when the measure space we are working over is clear. **sounds bad**

Lemma 2.2.1. Let φ be a simple function such that $\varphi = \sum_{i=1}^m a_i \chi_{E_i} = \sum_{j=1}^n b_j \chi_{F_j}$. If both of the families $\{E_i\}_{i=1}^m$ and $\{F_j\}_{j=1}^n$ are pairwise disjoint (respectively), then $\sum_{i=1}^m a_i \mu(E_i) = \sum_{j=1}^n b_j \mu(F_j)$.

Proof. For simplicity, define:

$$\begin{aligned} E_0 &:= X \setminus (E_1 \cup \dots \cup E_m), \\ F_0 &:= X \setminus (F_1 \cup \dots \cup F_n), \\ a_0 &:= 0, \\ b_0 &:= 0. \end{aligned}$$

Then:

$$\begin{aligned} \varphi &= \sum_{i=1}^m a_i \chi_{E_i} = \sum_{j=1}^n b_j \chi_{F_j} \\ &= \sum_{i=0}^m a_i \chi_{E_i} = \sum_{j=0}^n b_j \chi_{F_j}. \end{aligned}$$

Moreover, the families $\{E_i\}_{i=0}^m$ and $\{F_j\}_{j=0}^n$ are now partitions of X . Let $x \in X$ be arbitrary. Then there exists i, j such that $x \in E_i \cap F_j$. In particular:

$$\chi_{E_k}(x) = \begin{cases} 1 & k = i \\ 0 & k \neq i \end{cases}, \quad \chi_{F_\ell}(x) = \begin{cases} 1 & \ell = j \\ 0 & \ell \neq j \end{cases}.$$

Hence:

$$\varphi(x) = \sum_{i=0}^m a_i \chi_{E_i}(x) = a_i, \quad \varphi(x) = \sum_{j=0}^n b_j \chi_{F_j}(x) = b_j.$$

Since both series equal $\varphi(x)$, we must have $a_i = b_j$ on $E_i \cap F_j$. Lastly, note that for each i and j we can write:

$$\begin{aligned} E_i &= \bigcup_{j=0}^n (E_i \cap F_j), \\ F_j &= \bigcup_{i=0}^m (E_i \cap F_j). \end{aligned}$$

Having established all of the above facts, we can now see:

$$\begin{aligned}
\sum_{i=1}^m a_i \mu(E_i) &= \sum_{i=0}^m a_i \mu(E_i) \\
&= \sum_{i=0}^m a_i \mu\left(\bigcup_{j=0}^n E_i \cap F_j\right) \\
&= \sum_{i=0}^m \sum_{j=0}^n a_i \mu(E_i \cap F_j) \\
&= \sum_{j=0}^n \sum_{i=0}^m a_i \mu(E_i \cap F_j) \quad \text{Rearranging sum.} \\
&= \sum_{j=0}^n \sum_{i=0}^m b_j \mu(E_i \cap F_j) \\
&= \sum_{j=0}^n b_j \mu\left(\bigcup_{i=0}^m E_i \cap F_j\right) \\
&= \sum_{j=0}^n b_j \mu(F_j) \\
&= \sum_{j=1}^n b_j \mu(F_j). \quad \square
\end{aligned}$$

Proposition 2.2.2. *Let $\varphi, \psi \in S^+$.*

- (1) *If $c \geq 0$, then $\int c\varphi = c \int \varphi$.*
- (2) *If $\varphi \geq \psi$, then $\int \varphi \geq \int \psi$.*
- (3) *$\int(\varphi + \psi) = \int \varphi + \int \psi$.*
- (4) *For any fixed $A \in \mathcal{M}$, the map $A \mapsto \int_A \varphi$ defines a measure on $(X, \mathcal{M})$.*

Proof. (a) Let $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$ be in standard representation. We have:

$$\begin{aligned}
\int c\varphi &= \int c \sum_{i=1}^n a_i \chi_{E_i} \\
&= \int \sum_{i=1}^n c a_i \chi_{E_i} \\
&= \sum_{i=1}^n c a_i \mu(E_i) \\
&= c \sum_{i=1}^n a_i \mu(E_i) \\
&= c \int \varphi.
\end{aligned}$$

(b) Let $\varphi = \sum_{i=1}^m a_i \chi_{E_i}$ and $\psi = \sum_{j=1}^n b_j \chi_{F_j}$ be in standard representations. Note that we can again write $E_i = \bigcup_{j=0}^n (E_i \cap F_j)$ and $F_j = \bigcup_{i=0}^m (E_i \cap F_j)$ for all i, j . Hence on $E_i \cap F_j$, if $E_i \cap F_j = \emptyset$, then $\mu(E_i \cap F_j) = 0$. It must be the case that on $E_i \cap F_j$, since $\varphi \leq \psi$ we have $a_i \leq b_j$. This gives:

$$\begin{aligned} \int \varphi &= \sum_{i=1}^n a_i \mu(E_i) \\ &= \sum_{i=1}^n \sum_{j=1}^m a_i \mu(E_i \cap F_j) \\ &\leq \sum_{i=1}^n \sum_{j=1}^m b_j \mu(E_i \cap F_j) \\ &= \sum_{j=1}^m b_j \mu(F_j) \\ &= \int \psi. \end{aligned}$$

(c) need to eventually finish □

Proposition 2.2.3. Let $f \in L^+$.

(1) If $c \geq 0$, then $\int cf = c \int f$.

(2) If $f \leq g$, then $\int f \leq \int g$.

Proof. (a) If $c = 0$, then there is nothing to show. For $c > 0$:

$$\begin{aligned} \int f &= \sup \left\{ \int \varphi d\mu \mid 0 \leq \varphi \leq f, \varphi \in S^+ \right\} \\ &= \sup \left\{ \frac{1}{c} \cdot c \int \varphi d\mu \mid 0 \leq \varphi \leq f, \varphi \in S^+ \right\} \\ &= \frac{1}{c} \sup \left\{ c \int \varphi d\mu \mid 0 \leq \varphi \leq f, \varphi \in S^+ \right\} \\ &= \frac{1}{c} \sup \left\{ \int c\varphi d\mu \mid 0 \leq c\varphi \leq cf, \varphi \in S^+ \right\} \\ &= \frac{1}{c} \int cf. \end{aligned}$$

Multiplying both sides by c gives the desired result.

(b) Note that $\left\{ \int \varphi d\mu \mid 0 \leq \varphi \leq f, \varphi \in S^+ \right\} \subset \left\{ \int \varphi d\mu \mid 0 \leq \varphi \leq g, \varphi \in S^+ \right\}$. Therefore:

$$\sup \left\{ \int \varphi d\mu \mid 0 \leq \varphi \leq f, \varphi \in S^+ \right\} \leq \sup \left\{ \int \varphi d\mu \mid 0 \leq \varphi \leq g, \varphi \in S^+ \right\}$$

Hence $\int f \leq \int g$. □

In order to prove the equivalent (c) and (d) of Proposition ??, we need extra machinery.

Proposition 2.2.4. Let $f \in L^+$. Then $\int f = 0$ if and only if $f = 0$ a.e.

Proof. ($\Leftarrow$) Define $E := \{x \mid f(x) = 0\}$. Then by assumption $\mu(E^c) = 0$. Let $\varphi \in S^+$ and $\varphi \leq f$. Write $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$ in standard representation. Since $\varphi \leq f$...

Define $E := \{x \mid f(x) = 0\}$ and $N := E^c = \{x \mid f(x) \neq 0\}$. ($\Leftarrow$) Suppose $\mu(N) = 0$. Let $\varphi \in S^+$ and $0 \leq \varphi \leq f$. Suppose $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$ is in standard representation. Since $\varphi \leq f$, if $a_i > 0$ then it must be the case that $E_i \subset N^*$ (we know that $\varphi \leq f$, where $f = 0$ a.e., so if $a_i > 0$, it must be the case that $\mu(E_i) = 0$). Hence $\mu(E_i) = 0$. But since $\int \varphi = \sum_{i=1}^n a_i \mu(E_i) = 0$, it must be the case that $\sup\{\int \varphi \mid 0 \leq \varphi \leq f, \varphi \in S^+\} = 0$.

($\Rightarrow$) Now suppose $\int f = 0$. Let $E_n = \{x \mid f(x) \geq \frac{1}{n}\}$. Then $E_1 \subset E_2 \subset E_3 \subset \dots$, and $P := \bigcup_{i=1}^{\infty} E_i = \{x \mid f(x) > 0\}$. If $\mu(P) = 0$, then we are done. If $\mu(P) > 0$, then since $\mu(P) = \lim_{n \rightarrow \infty} \mu(E_n)$, there must exist some $n \in \mathbb{N}$ such that $\mu(E_n) > 0$. It follows that:

$$\begin{aligned} \int f &\geq \int_{E_n} f \\ &= \int f \chi_{E_n} \\ &\geq \int \frac{1}{n} \chi_{E_n} \\ &= \frac{1}{n} \mu(E_n) \\ &> 0. \end{aligned}$$

This is a contradiction, so it must be the case that $\mu(P) = 0$. Hence $f = 0$ a.e. $\square$

Theorem 2.2.2 (Monotone Convergence Theorem). *Let $(f_n)_n$ be an increasing sequence in L^+ , and suppose $\lim_{n \rightarrow \infty} f_n = f$. Then $\int f = \lim_{n \rightarrow \infty} \int f_n$.*

Proof. We will prove this by showing inequality in both directions. Note that for all $n \in \mathbb{N}$ we have $f_n \leq f$. It must be the case that if $\varphi \in S^+$ such that $\varphi \leq f_n$, then $\varphi \leq f$ as well. We've obtained the set inclusion:

$$\left\{ \int \varphi \mid \varphi \leq f_n, \varphi \in S^+ \right\} \subset \left\{ \int \varphi \mid \varphi \leq f, \varphi \in S^+ \right\}.$$

Hence:

$$\begin{aligned} \int f &= \sup \left\{ \int \varphi \mid \varphi \leq f, \varphi \in S^+ \right\} \\ &\geq \sup \left\{ \int \varphi \mid \varphi \leq f_n, \varphi \in S^+ \right\} \\ &= \int f_n. \end{aligned}$$

Taking $n \rightarrow \infty$ gives $\int f \geq \lim_{n \rightarrow \infty} \int f_n$.

Now fix $\varphi \in S^+$ such that $0 \leq \varphi \leq f$, and fix a constant $0 < \alpha < 1$. Define $E_n := \{x \mid f_n(x) \geq \alpha \varphi(x)\}$. We can see that:

$$\begin{aligned} E_n &= \{x \mid f_n(x) \geq \alpha \varphi(x)\} \\ &= \{x \mid (f_n - \alpha \varphi)(x) \geq 0\} \\ &= \{x \mid (f_n - \alpha \varphi)(x) \in [0, \infty]\} \\ &= (f_n - \alpha \varphi)^{-1}([0, \infty]) \end{aligned}$$

Since f_n are φ are $\mathcal{M}$ -measurable functions, their difference $f_n - \alpha\varphi$ is $\mathcal{M}$ -measurable, hence $E_n \in \mathcal{M}$. In particular, since $(f_n)_n$ is an increasing sequence, we have $E_1 \subset E_2 \subset E_3 \subset \dots$ and $\bigcup_{i=1}^{\infty} E_i = X^*$ ($(f_n)_n$ is an increasing sequence so surely $f_{n+1} \geq f_n(x) \geq \alpha\varphi(x)$, hence $E_{n+1} \supset E_n$. also, $\varphi \leq f$, and since $\alpha \in (0, 1)$, we have $\alpha\varphi < f$. but this is true for all x , and $f_n \rightarrow f$, hence the unions of all the E 's should be our entire set.). Now note that for all $x \in X$ we have $\alpha\varphi\chi_{E_n} \leq f_n$, which gives:

$$\begin{aligned} \int f_n &\geq \int \alpha\varphi\chi_{E_n} \\ &= \alpha \int_{E_n} \varphi. \end{aligned}$$

But recall that $A \mapsto \int_A \varphi$ defines a measure on $(X, \mathcal{M})$! Applying continuity from below gives:

$$\begin{aligned} \alpha \int \varphi &= \lim_{n \rightarrow \infty} \alpha \int_{E_n} \varphi \\ &\leq \lim_{n \rightarrow \infty} \int f_n. \end{aligned}$$

Sending $\alpha \rightarrow 1$ gives $\int \varphi \leq \lim_{n \rightarrow \infty} \int f_n$. Taking the supremum over all $\varphi \in S^+$ such that $\varphi \leq f$ gives $\int f \leq \lim_{n \rightarrow \infty} \int f_n$. $\square$

Example 2.2.1. *do the examples of how it fails to hold for nonincreasing functions*

Corollary 2.2.1. *Let $f, g \in L^+$. We have $\int(f + g) = \int f + \int g$.*

Proof. By Proposition 2.2.1 we can find increasing sequences of simple functions $(\varphi_n)_n$ and $(\psi_n)_n$ such that $\varphi_n \nearrow f$ and $\psi_n \nearrow g$. Since each sequence is increasing and convergent, we have $\varphi_n + \psi_n \nearrow f + g$. By the Monotone Convergence Theorem:

$$\begin{aligned} \int(f + g) &= \lim_{n \rightarrow \infty} \int(\varphi_n + \psi_n) \\ &= \lim_{n \rightarrow \infty} \int \varphi_n + \lim_{n \rightarrow \infty} \int \psi_n \\ &= \int f + \int g. \end{aligned} \quad \square$$

Definition 2.2.5. *Let $(f_n)_n$ be a function in L^+ . We say $(f_n)_n$ converges almost everywhere if the set:*

$$\{x \mid f_n(x) \not\rightarrow f(x)\}$$

has measure zero, and denote this as $f_n \rightarrow f$ a.e.

We can use the Monotone Convergence Theorem to prove a more general result on increasing sequence of functions which converge almost everywhere.

Theorem 2.2.3. *Let $(f_n)_n$ be a sequence in L^+ and suppose $f_n \nearrow f$ a.e. for some $f \in L^+$. Then $\int f = \lim_{n \rightarrow \infty} \int f_n$.*

Proof. Define $E := \{x \mid f_n(x) \nearrow f(x)\}$. Then by assumption E^c is of measure zero, so $f_n \chi_E \nearrow f \chi_E$ holds everywhere. By the Monotone Convergence Theorem, we have:

$$\int f \chi_E = \lim_{n \rightarrow \infty} \int f_n \chi_E.$$

Note that $f = f \chi_E$ a.e., hence $f - f \chi_E = 0$ a.e. By Proposition 2.1.5 we have that $\int (f - f \chi_E) = 0$, hence $\int f = \int f \chi_E$. Similarly, $f_n = f_n \chi_E$ a.e., hence by the same argument we have $\int f_n = \int f_n \chi_E$. Together, $\int f = \lim_{n \rightarrow \infty} \int f_n$. $\square$

Theorem 2.2.4 (Fatou's Lemma). *Let $(f_n)_n$ be a sequence in L^+ . Then $\int \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int f_n$.*

Proof. For each $k \in \mathbb{N}$, define $g_n := \inf_{k \geq n} f_k$. Then g_n is measurable by Proposition 2.1.5, and in particular $g_n \nearrow \liminf_{n \rightarrow \infty} f_n$ (since the limit infimums form an increasing sequence). Note that $g_n \leq f_k$ for all $k \geq n$, hence $\int g_n \leq \int f_k$ for all $k \geq n$. Since $\int g_n$ is a lowerbound for the set $\{\int f_k \mid k \geq n\}$, it must be the case that $\int g_n \leq \inf_{k \geq n} \int f_k$. By the Monotone Convergence Theorem:

$$\begin{aligned} \int \liminf_{n \rightarrow \infty} f_n &\leq \lim_{n \rightarrow \infty} \int g_n \\ &\leq \lim_{n \rightarrow \infty} \left(\inf_{k \geq n} \int f_k \right) \\ &= \liminf_{n \rightarrow \infty} \int f_n. \end{aligned} \quad \square$$

Corollary 2.2.2. *Let $(f_n)_n$ be a sequence in L^+ . Suppose $f_n \rightarrow f$ a.e. Then $\int f \leq \liminf_{n \rightarrow \infty} \int f_n$.*

Proof. sketch: let $E = \{x \mid f_n(x) \rightarrow f(x)\}$. Replace f w/ $f \chi_E$ and f_n w/ $f_n \chi_E$. apply fatous lemma i think. $\square$

Definition 2.2.6. *Let $f : X \rightarrow \overline{\mathbb{R}}$ be measurable.*

- (1) *We say f is integrable if $\int |f| < \infty$.*
- (2) *If $E \in \mathcal{M}$, f is integrable on E if $\int_E |f| < \infty$.*

Proposition 2.2.5. *$f : X \rightarrow \mathbb{C}$ is integrable if and only if $\Re(f)$ and $\Im(f)$ are integrable.*

not sure how to do this rigorously

Proof. $\square$

Proposition 2.2.6. *Let $f : X \rightarrow \mathbb{C}$ be integrable.*

- (1) *For all $\alpha \in \mathbb{C}$, $\int \alpha f = \alpha \int f$.*
- (2) *$|\int f| \leq \int |f|$.*

Proof. (1) Write:

$$\begin{aligned} \alpha &= a + ib, \quad a, b \in \mathbb{R}; \\ f &= u + iv, \quad u, v : X \rightarrow \mathbb{R}. \end{aligned}$$

Then:

$$\begin{aligned}
 \int \alpha f &= \int (a + ib)(u + iv) \\
 &= \int (au + ibu + iav - bv) \\
 &= \int (au + ibu + v(ia - b)) \\
 &= \int ((a + ib)u + vi(a + ib)) \\
 &= (a + ib) \int u + (a + ib)i \int v \\
 &= (a + ib) \left(\int u + i \int v \right) \\
 &= (a + ib) \left(\int u + iv \right) \\
 &= \alpha \int f.
 \end{aligned}$$

(2) If $\int f = 0$, we are done. Otherwise, let $\alpha := \frac{\int f}{\int f} \in \mathbb{C}$ Then:

$$\begin{aligned}
 \left| \int f \right| &= \alpha \int f \\
 &= \int \alpha f \\
 &= \int \Re(\alpha f) + i \int \Im(\alpha f) \\
 &\leq \int |\alpha f| \\
 &= \int |f|.
 \end{aligned}$$

□

Proposition 2.2.7. *Let $f : X \rightarrow \mathbb{C}$ be integrable.*

- (1) $\{x \mid f(x) = \pm\infty\}$ is a null set.
- (2) There exists $g : X \rightarrow \mathbb{C}$ such that $f = g$ a.e.

Proof. **sadge**

□

Theorem 2.2.5. *Equality almost everywhere defines an equivalence relation on the set $\{f : X \rightarrow \mathbb{C} \mid f \text{ is integrable}\}$.*

Proof. Given $f, g \in \{f : X \rightarrow \mathbb{C} \mid f \text{ is integrable}\}$, write $f \sim g$ if $f = g$ a.e. Clearly $\sim$ is symmetric and reflexive, so we only need to show that it is transitive. Suppose $f = g$ a.e. and $g = h$ a.e. Let $E_1 = \{x \mid f(x) \neq g(x)\}$ and $E_2 = \{x \mid g(x) \neq h(x)\}$. By hypothesis $\mu(E_1^c) = \mu(E_2^c) = 0$. Now note that $f = g\chi_{E_1}$ and $g = h\chi_{E_2}$ everywhere. Multiplying both sides of the second equation by χ_{E_1} gives $g\chi_{E_1} = h\chi_{E_1}\chi_{E_2} = h\chi_{E_1 \cap E_2}$. Together, we have $f = h\chi_{E_1 \cap E_2}$ everywhere. But note that $\mu((E_1 \cap E_2)^c) = \mu(E_1^c \cup E_2^c) \leq \mu(E_1^c) + \mu(E_2^c) = 0$. Hence $f = h$ a.e., completing the proof. □

Definition 2.2.7.

- (1) The Lebesgue space $L^1(X, \mu)$ is the set of equivalence classes of integrable functions $f : X \rightarrow \mathbb{C}$ modulo equality almost everywhere; i.e., using the terminology in Theorem 2.2.5:

$$L^1(X, \mu) := \{f : X \rightarrow \mathbb{C} \mid f \text{ is integrable}\} / \sim.$$

- (2) Given $f \in L^1(X, \mu)$, the L^1 -norm of f is:

$$\|f\|_{L^1} := \int |f|.$$

Theorem 2.2.6. $(L^1(X, \mu), \|\cdot\|_{L^1})$ is a normed vector space over $\mathbb{C}$.

Proof. **hard, might just show that addition and scalar multipliacion are well-defined, and say that it follows from the fact that the pre-quotient space is a complex vector space.** $\square$

Remark. If context is clear, we write L^1 instead of $L^1(X, \mu)$. Moreover, it is normal to abuse notation and speak of a "function" in L^1 rather than an equivalence class (which is technically more correct).

Theorem 2.2.7 (Dominated Convergence Theorem). *Let $(f_n)_n$ be a sequence in L^1 and suppose $f_n \rightarrow f$ a.e. for some $f : X \rightarrow \mathbb{C}$. Moreover, suppose there exists $g \in L^1$ such that $g \geq 0$ and for all $n \in \mathbb{N}$, we have $|f_n| \leq g$ a.e. Then (after possibly changing f on a null set), $f \in L^1$ and $\int f = \lim_{n \rightarrow \infty} \int f_n$.*

Proof. Define $E_0 := \{x \mid f_n(x) \rightarrow f(x)\}$ and for $n \geq 1$ define $E_n := \{x \mid |f_n(x)| \leq g(x)\}$. Note that for each $i \in \mathbb{N}$, the measure of E_i^c is zero by assumption. Now let $E := \bigcap_{n=0}^{\infty} E_n$. We can see:

$$\begin{aligned} \mu(E^c) &= \mu\left(\left(\bigcap_{n=0}^{\infty} E_n\right)^c\right) \\ &= \mu\left(\bigcup_{n=0}^{\infty} E_n^c\right) \\ &\leq \sum_{n=0}^{\infty} \mu(E_n^c) \\ &= 0. \end{aligned}$$

Hence $f_n \chi_E \rightarrow f \chi_E$ everywhere, and in fact $f \chi_E$ is measurable by Proposition 2.1.5. Moreover, for each $n \in \mathbb{N}$ we have $|f_n \chi_E| \leq g \chi_E \leq g$. Together:

$$\int |f_n \chi_E| \leq \int g < \infty.$$

Hence:

$$\int |f \chi_E| \leq \sup_{n \in \mathbb{N}} \int |f_n \chi_E| \leq \int g < \infty.$$

Thus $f \chi_E \in L^1$. Now write $f_n \chi_E = u_n + i v_n$ and $f \chi_E = u + i v$, where $u_n \rightarrow u$ and $v_n \rightarrow v$. Surely $g \pm u_n \geq 0$ and $g \pm v_n \geq 0$, hence by Fatou's Lemma:

$$\int \liminf_{n \rightarrow \infty} (g + u_n) \leq \liminf_{n \rightarrow \infty} \int (g + u_n).$$

Since $u_n \rightarrow u$ we have $\liminf_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} u_n = u$, giving:

$$\int g + \int u \leq \int g + \liminf_{n \rightarrow \infty} \int u_n.$$

Subtracting $\int g$ from both sides gives $\int u \leq \liminf_{n \rightarrow \infty} \int u_n$. Now applying Fatou's Lemma to $g - u_n$ gives:

$$\int \liminf_{n \rightarrow \infty} (g - u_n) \leq \liminf_{n \rightarrow \infty} \int (g - u_n).$$

Hence:

$$\begin{aligned} \int g - \int u &\leq \int g + \liminf_{n \rightarrow \infty} \left(\int -u_n \right) \\ &= \int g - \limsup_{n \rightarrow \infty} \int u_n. \end{aligned}$$

Subtracting $\int g$ from both sides and rearranging gives $\int u \geq \limsup_{n \rightarrow \infty} \int u_n$. Together:

$$\limsup_{n \rightarrow \infty} \int u_n \leq \int u \leq \liminf_{n \rightarrow \infty} \int u_n.$$

But since $\liminf_{n \rightarrow \infty} \int u_n \leq \int u \leq \limsup_{n \rightarrow \infty} \int u_n$ trivially, we have:

$$\limsup_{n \rightarrow \infty} \int u_n = \int u = \liminf_{n \rightarrow \infty} \int u_n.$$

Thus $\lim_{n \rightarrow \infty} \int u_n = \int u$. An identical argument gives $\lim_{n \rightarrow \infty} \int v_n = \int v$. Together:

$$\begin{aligned} \int f\chi_E &= \int (u + iv) \\ &= \int u + i \int v \\ &= \lim_{n \rightarrow \infty} \int u_n + i \lim_{n \rightarrow \infty} \int v_n \\ &= \lim_{n \rightarrow \infty} \left(\int u_n + i \int v_n \right) \\ &= \lim_{n \rightarrow \infty} \int (u_n + iv_n) \\ &= \lim_{n \rightarrow \infty} \int f_n \chi_E. \end{aligned}$$

Note that $f\chi_E$ is precisely f changed on a null set. □

Theorem 2.2.8. *Let $(f_n)_n$ be a sequence in L^1 with $\sum_{n=1}^{\infty} \int |f_n| < \infty$. There exists $f \in L^1$ such that $\sum_{n=1}^N f_n \rightarrow f$ a.e. as $N \rightarrow \infty$ (i.e., $\sum_{n=1}^{\infty} f_n = f$ a.e.) and $\int f = \int \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int f_n$.*

Proof. Define $g_n := \sum_{j=1}^n |f_j|$ and $g := \sum_{j=1}^{\infty} |f_j|$. Note that g_n is measurable by Proposition 2.1.3, and g is measurable by hypothesis. Moreover, we have $0 \leq g_n$ and $g_n \nearrow g$. By the Monotone Convergence Theorem:

$$\begin{aligned} \int g &= \lim_{n \rightarrow \infty} \int f_n \\ &= \lim_{n \rightarrow \infty} \int \sum_{j=1}^n |f_j| \\ &= \sum_{j=1}^{\infty} \int |f_j| \\ &< \infty. \end{aligned}$$

Hence g is integrable. Define $E := \{x \mid g(x) < \infty\}$. By Proposition 2.2.7, the set $E^c = \{x \mid g(x) = \infty\}$ is a null set. In particular, $g\chi_E \leq g$, hence $g\chi_E \in L^1$. Now define:

$$f(x) := \begin{cases} \sum_{n=1}^{\infty} f_n(x), & x \in E \\ 0, & x \notin E. \end{cases}$$

We have that $\sum_{n=1}^{\infty} f_n = f$ a.e. Likewise:

$$\begin{aligned} \left| \sum_{j=1}^n f_j \right| &\leq \sum_{j=1}^n |f_j| \\ &\leq \sum_{j=1}^{\infty} |f_j| \\ &= g. \end{aligned}$$

Hence $\left| \sum_{j=1}^n f_j \right| \leq g\chi_E$ a.e., and $\sum_{j=1}^n f_j \rightarrow f$ a.e. By the Dominated Convergence Theorem, $f \in L^1$ and $\int f = \lim_{n \rightarrow \infty} \int \sum_{j=1}^n f_n$. Finally:

$$\begin{aligned} \int \sum_{n=1}^{\infty} f_n &= \int f \\ &= \lim_{n \rightarrow \infty} \int \sum_{j=1}^n f_n \\ &= \lim_{n \rightarrow \infty} \sum_{j=1}^n \int f_n \\ &= \sum_{n=1}^{\infty} \int f_n. \end{aligned} \quad \square$$

The ability to interchange integrals and infinite series is useful in application. More importantly however, we showed in Theorem 2.2.8 that *every absolutely convergent series is convergent*. That is, given a sequence of functions $(f_n)_n \subset L^1$, if $\sum_{n=1}^{\infty} \|f_n\|_{L^1}$ converges, then $\sum_{n=1}^{\infty} f_n$ converges. The next result is crucially dependent on this fact.

Theorem 2.2.9. $(L^1, \|\cdot\|_{L^1})$ is a Banach space.

Proof. We proved in Theorem 2.2.6 that $(L^1, \|\cdot\|_{L^1})$ is a normed space, so it only remains to show that this space is complete. Let $(f_n)_n$ be $\|\cdot\|_{L^1}$ -Cauchy. Find $n_1 \in \mathbb{N}$ such that $p, q \geq n_1$ implies $\|f_p - f_q\|_{L^1} < \frac{1}{2}$. Find $n_2 \in \mathbb{N}$ such that $n_2 > n_1$ and $p, q \geq n_2$ implies $\|f_p - f_q\|_{L^1} < \frac{1}{2^2}$. Inductively, find $n_k \in \mathbb{N}$ such that $n_k > n_{k-1}$ and $p, q \geq n_k$ implies $\|f_p - f_q\|_{L^1} < \frac{1}{2^k}$. Consider the sequence $(f_{n_{k+1}} - f_{n_k})_k$. Then:

$$\sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\| \leq \sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$$

Since $\sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|$ converges, by Theorem 2.2.8 there exists $f \in L^1$ such that $\sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k}) = f$. Examining the partial sums of $\sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$, we can see that:

$$\sum_{k=1}^m (f_{n_{k+1}} - f_{n_k}) = f_{n_{m+1}} - f_{n_1}.$$

Hence $\lim_{m \rightarrow \infty} (f_{n_{m+1}} - f_{n_1}) = f$. Finally:

$$\begin{aligned} \lim_{m \rightarrow \infty} f_{n_m} &= \lim_{m \rightarrow \infty} (f_{n_m} - f_{n_1} + f_{n_1}) \\ &= \lim_{m \rightarrow \infty} (f_{n_m} - f_{n_1}) + \lim_{m \rightarrow \infty} f_{n_1} \\ &= f + f_{n_1} \in L^1. \end{aligned}$$

Since $(f_n)_n$ is a $\|\cdot\|_{L^1}$ -Cauchy sequence which admits a convergent subsequence, $(f_n)_n$ converges. Thus $(L^1, \|\cdot\|_{L^1})$ is a Banach space. $\square$

Theorem 2.2.10. The set $\{\varphi \in L^1 \mid \varphi \text{ is simple}\}$ is dense in L^1 .

Proof. Let $f \in L^1$ and let $\epsilon > 0$. We proceed by cases on f .

Case 1: $f \geq 0$. By Theorem 2.2.1, there exists a sequence $(\varphi_n)_n$ of simple functions such that $\varphi_n \nearrow f$. By the Monotone Convergence Theorem, $\int f = \lim_{n \rightarrow \infty} \int \varphi_n$. Find $N \in \mathbb{N}$ sufficiently large so that $|\int f - \int \varphi| < \epsilon$. Since $f - \varphi \geq 0$, we have $|\int f - \int \varphi| = \int (f - \varphi) = \int |f - \varphi|$. Hence $\int |f - \varphi| < \epsilon$; i.e., $\|f - \varphi\|_{L^1} < \epsilon$.

Case 2: $f \in \mathbb{C}$. Write:

$$\begin{aligned} f &= u + iv \\ &= (u^+ - u^-) + i(v^+ - v^-). \end{aligned}$$

From Case 1, choose simple functions $\varphi^+, \varphi^-, \psi^+, \psi^-$ such that:

$$\begin{aligned} \|\varphi^+ - u^+\|_{L^1} &< \frac{\epsilon}{4}, \\ \|\varphi^- - u^-\|_{L^1} &< \frac{\epsilon}{4}, \\ \|\psi^+ - v^+\|_{L^1} &< \frac{\epsilon}{4}, \\ \|\psi^- - v^-\|_{L^1} &< \frac{\epsilon}{4}. \end{aligned}$$

Define $\xi := (\varphi^+ - \varphi^-) + i(\psi^+ - \psi^-)$. Then:

$$\begin{aligned} \|f - \xi\|_{L^1} &= \|((u^+ - u^-) + i(v^+ - v^-)) - ((\varphi^+ - \varphi^-) + i(\psi^+ - \psi^-))\| \\ &\leq \|u^+ - \varphi^+\|_{L^1} + \|u^- - \varphi^-\|_{L^1} + \|i(v^+ - \psi^+)\|_{L^1} + \|i(v^- - \psi^-)\|_{L^1} \\ &< \frac{\epsilon}{4} + \frac{\epsilon}{4} + \frac{\epsilon}{4} + \frac{\epsilon}{4} \\ &= \epsilon. \end{aligned}$$

Thus $\{\varphi \in L^1 \mid \varphi \text{ is simple}\}$ is dense in L^1 . □

§ 2.3. Lebesgue Integration on $\mathbb{R}$

Let $(X, \mathcal{M}, \mu) = (\mathbb{R}, \mathcal{L}, m)$. **recall this stuff: talk about the riemann integral with pictures and stuff**

Definition 2.3.1. Let $[a, b] \subseteq \mathbb{R}$.

(1) A partition P of the interval $[a, b]$ is a finite set of points $\{x_0, \dots, x_n\} \subseteq [a, b]$ such that:

$$a = x_0 < x_1 < \dots, x_{n-1} < x_n = b.$$

(2) A refinement of a partition P is a partition Q such that $P \subset Q$. In this case, we say that Q is finer than P . Given two partitions $P_1, P_2 \subseteq [a, b]$, we call the union $P_1 \cup P_2$ their common refinement.

(3) The change in x_i is denoted $\Delta x_i := x_i - x_{i-1}$.

(4) The mesh of a partition P is denoted $\|P\| := \max_{i=1}^n \Delta x_i$.

Definition 2.3.2. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function and $P = \{x_0, \dots, x_n\}$ a partition of $[a, b]$. We define:

(1) $M_j(f) = \sup_{x \in [x_{j-1}, x_j]} f(x)$; and

(2) $m_j(f) = \inf_{x \in [x_{j-1}, x_j]} f(x)$.

We call the numbers:

$$S_P(f) = \sum_{j=1}^n M_j(f) \Delta x_j, \quad s_P(f) = \sum_{j=1}^n m_j(f) \Delta x_j,$$

the upper Darboux sum and, respectively, the lower Darboux sum of f over the partition P .

Definition 2.3.3. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function.

(1) The upper Darboux integral of f over $[a, b]$ is:

$$\overline{I}_a^b(f) := \inf_{P \in \mathcal{B}_{[a,b]}} U(P, f).$$

(2) The lower Darboux integral of f over $[a, b]$ is:

$$\underline{I}_a^b(f) := \sup_{P \in \mathcal{B}_{[a,b]}} L(P, f).$$

(3) We say that f is Riemann integrable on $[a, b]$ provided that:

$$\overline{I}_a^b(f) = \underline{I}_a^b(f).$$

In this case, the common value of the upper and lower Darboux integrals is called the Riemann integral of f over $[a, b]$ and is denoted:

$$\int_a^b f(x)dx.$$

Theorem 2.3.1. Let f be a bounded real function on $[a, b]$. If f is Riemann integrable, then f is Lebesgue measurable and $\int_a^b f(x)dx = \int_{[a,b]} f d\mu$.

Proof. For any partition $P = \{x_0, x_1, \dots, x_n\}$, define:

$$G_P := \sum_{i=1}^n M_i(f) \chi_{(x_{i-1}, x_i]},$$

$$g_P := \sum_{i=1}^n m_i(f) \chi_{(x_{i-1}, x_i]}.$$

Note that $g_P \leq f \leq G_P$ holds for any partition P . Choose a sequence of partitions $P_1 \subset P_2 \subset P_3 \subset \dots$ such that:

$$\int_{[a,b]} G_{P_k} d\mu = S_{P_k}(f) \searrow \int_a^b f(x)dx$$

$$\int_{[a,b]} g_{P_k} d\mu = s_{P_k}(f) \nearrow \int_a^b f(x)dx.$$

Let $G := \lim_{k \rightarrow \infty} G_{P_k}$ and $g := \lim_{k \rightarrow \infty} g_{P_k}$. Note that g and G are measurable since they are limits of measurable functions. Moreover, by construction $g \leq f \leq G$, hence $G - g \in L^+$. Applying the Dominated Convergence Theorem gives:

$$\int_{[a,b]} G d\mu = \lim_{k \rightarrow \infty} \int_{[a,b]} G_{P_k} d\mu = \int_a^b f(x)dx$$

$$\int_{[a,b]} g d\mu = \lim_{k \rightarrow \infty} \int_{[a,b]} g_{P_k} d\mu = \int_a^b f(x)dx.$$

Observe that:

$$\begin{aligned} \int_{[a,b]} (G - g) d\mu &= \int_{[a,b]} G d\mu - \int_{[a,b]} g d\mu \\ &= \int_a^b f(x)dx - \int_a^b f(x)dx \\ &= 0. \end{aligned}$$

By Proposition 2.2.4, $G - g = 0$ a.e., hence $G = g$ a.e. Since f is squeezed between g and G , it must be the case that $f = G$ a.e. (or similarly, $f = g$ a.e.).

Claim: Since G is measurable and since m is complete, f is Lebesgue measurable. Define $E := \{x \mid G(x) = g(x)\}$. Then $E = (G - g)^{-1}(\{0\})$, hence $E \in \mathcal{L}$. Define $N := E^c$, then by hypothesis $m(N) = 0$. To show f is measurable, we will make use of (2) in Proposition 2.1.1. Let $U \in \tau_{\mathbb{R}}$. We have:

$$\begin{aligned} f^{-1}(U) &= (f^{-1}(U) \cap E) \cup (f^{-1}(U) \cap N) \\ &= (G^{-1}(U) \cap E) \cup (f^{-1}(U) \cap N) \quad \text{Since } f = G \text{ on } E. \end{aligned}$$

Note that $G^{-1}(U) \in \mathcal{L}$, hence $G^{-1}(U) \cap E \in \mathcal{L}$. Moreover, $f^{-1}(U) \cap N \subset N$, and since m is complete it must be the case that $f^{-1}(U) \cap N \in \mathcal{L}$. Thus $f^{-1}(U) \in \mathcal{L}$, which means f is measurable. This implies that $G - f \in L^+$, and since $G = f$ a.e., applying Proposition 2.2.4 again gives:

$$\begin{aligned} \int_{[a,b]} f dm &= \int [a,b] G dm \\ &= \int_a^b f(x) dx. \end{aligned}$$

double check this later when you aren't geeked. □

—————//—————

Definition 2.3.4. Let $f : X \rightarrow \mathbb{R}$. The support of f is $\text{supp}(f) = \{x \mid f(x) \neq 0\}$.

Definition 2.3.5.

- (1) $C^0(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is continuous}\}$.
- (2) $C^n(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is } n\text{-times continuously differentiable}\}$.
- (3) $C^\infty(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is smooth}\}$.
- (4) $C_c^0(\mathbb{R}) = \{f \in C^0(\mathbb{R}) \mid \text{supp}(f) \text{ is compact}\}$.

Lemma 2.3.1. Let $I = (a, b)$ be an interval where $-\infty < a < b < \infty$. For all $\epsilon > 0$, there exists $f \in C_c^0(\mathbb{R})$ such that $\text{supp}(f) \subset I$ and $\|\chi_I - f\|_{L^1} < \epsilon$.

Proof. For each $n \in \mathbb{N}$, define $\delta_n := \frac{b-a}{2n}$ and:

$$f_n(x) := \begin{cases} 0, & x \leq a, x \geq b \\ 1, & a + \delta_n \leq x, x \leq b - \delta_n \\ \frac{x-a}{\delta_n}, & a \leq x \leq a + \delta_n \\ \frac{b-x}{\delta_n}, & b - \delta_n \leq x \leq b. \end{cases}$$

Note that $\chi_I \geq f_n$ for all $n \in \mathbb{N}$, hence:

$$\begin{aligned} \|\chi_I - f_n\|_{L^1} &= \int |\chi_I - f_n| \\ &= \frac{1}{2}\delta_n \cdot 1 + \frac{1}{2}\delta_n \cdot 1 \\ &= \delta_n. \end{aligned}$$

Take n to be large enough such that $\delta_n < \epsilon$. □

Theorem 2.3.2. $C_c^0(\mathbb{R})$ is dense in $L^1(\mathbb{R}, m)$.

Proof. Let $g \in L^1(\mathbb{R}, m)$ and $\epsilon > 0$. By Theorem 2.2.10, find a simple function $\varphi \in L^1(\mathbb{R}, m)$ such that $\|g - \varphi\|_{L^1} < \frac{\epsilon}{3}$. Let $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$ be in standard form (that is, $a_i = 0$ for each i and $E_i \cap E_j = \emptyset$ for each $i \neq j$). By Proposition 1.4.4, for each $i \in \mathbb{N}$ find an open set U_i which is the finite union of bounded open intervals such that:

$$m(E_i \Delta U_i) < \frac{\epsilon}{3n|a_i|}.$$

Let $\Psi = \sum_{i=1}^n a_i \chi_{U_i}$. Then:

$$\begin{aligned} \|\varphi - \Psi\|_{L^1} &= \left\| \sum_{i=1}^n a_i \chi_{E_i} - \sum_{i=1}^n a_i \chi_{U_i} \right\|_{L^1} \\ &= \left\| \sum_{i=1}^n a_i (\chi_{E_i} - \chi_{U_i}) \right\|_{L^1} \\ &\leq \sum_{i=1}^n \|a_i \chi_{E_i} - a_i \chi_{U_i}\|_{L^1} \\ &\leq \sum_{i=1}^n \|a_i \chi_{E_i \Delta U_i}\|_{L^1} \\ &= \sum_{i=1}^n |a_i| m(E_i \Delta U_i) \\ &< \sum_{i=1}^n \frac{\epsilon}{3n} \\ &= \frac{\epsilon}{3}. \end{aligned}$$

Now for each $i \in \mathbb{N}$, denote $U_i = \bigcup_{j=1}^{N_i} I_{i,j}$, where each $I_{i,j}$ is a bounded open interval. By Lemma 2.3.1, for every i and j choose $f_{i,j} \in C^0(\mathbb{R})$ such that $\text{supp}(f_{i,j}) \subset I_{i,j}$ and:

$$\|\chi_{I_{i,j}} - f_{i,j}\|_{L^1} < \frac{\epsilon}{3N_i|a_i|n}.$$

Define $f := \sum_{i=1}^n \sum_{j=1}^{N_i} |a_i| f_{i,j}$. Note that f is continuous since it is a finite linear combination of continuous functions. Moreover, f is supported on the bounded set $\bigcup_{i,j} I_{i,j}$. $\square$

§ 2.4. Modes of Convergence

Definition 2.4.1. Let $(f_n)_n$ be a sequence of functions in L^+ and let $f \in L^+$.

- (1) $f_n \rightarrow f$ uniformly if $\sup_{x \in X} |f(x) - f_n(x)| \rightarrow 0$.
- (2) $f_n \rightarrow f$ in L^1 if $\|f - f_n\|_{L^1} \rightarrow 0$.
- (3) $f_n \rightarrow f$ in measure if for all $\epsilon > 0$, $\mu(\{x \mid |f_n(x) - f(x)| \geq \epsilon\}) \rightarrow 0$.
- (4) f_n is Cauchy in measure if for all $\epsilon > 0$, $\mu(\{x \mid |f_n(x) - f_m(x)| \geq \epsilon\}) \rightarrow 0$.

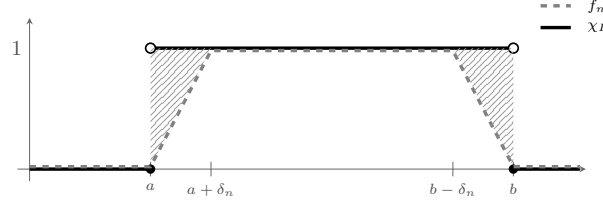


Figure 2.1: Illustration of f_n and χ_I . Note that the hatched area represents $\|\chi_I - f_n\|_{L^1}$.

Example 2.4.1. *examples and pictures go here*

Proposition 2.4.1. *If $f_n \rightarrow f$ in L^1 , then $f_n \rightarrow f$ in measure.*

Proof. Let $\epsilon > 0$. For all $n \in \mathbb{N}$, define $E_n := \{x \mid |f_n(x) - f(x)| \geq \epsilon\}$. We have:

$$\begin{aligned} \|f - f_n\|_{L^1} &= \int |f - f_n| \\ &\geq \int_{E_n} |f - f_n| \\ &\geq \int_{E_n} \epsilon \\ &= \epsilon \cdot \mu(E_n). \end{aligned}$$

Hence $\mu(E_n) \leq \frac{1}{\epsilon} \|f - f_n\|_{L^1}$. Since $\|f - f_n\|_{L^1} \rightarrow 0$ as $n \rightarrow \infty$, we have $\mu(E_n) \rightarrow 0$. $\square$

Theorem 2.4.1 (Egorov's Theorem). *Let $\mu(X) < \infty$. For each $n \in \mathbb{N}_0$, suppose f_n is a measurable complex-valued function such that $f_n \rightarrow f_0$ pointwise almost everywhere. Then for all $\epsilon > 0$, there exists a measurable set $E \in \mathcal{M}$ such that $\mu(E) < \epsilon$ and $f_n \rightarrow f$ uniformly on E^c .*

Proof. Let $\epsilon > 0$. Define $A := \{x \mid f_n(x) \rightarrow f_0(x)\}$. By hypothesis $\mu(A^c) = 0$. For all $n, k \in \mathbb{N}$, define:

$$E_n(k) = \bigcup_{m \geq n} \left\{ x \in A \mid |f_m(x) - f(x)| \geq \frac{1}{k} \right\}.$$

For a fixed k , we have $E_n(k) \supset E_{n+1}(k)$ for all n , and $\bigcap_{n=1}^{\infty} E_n(k) = \emptyset$. Since $\mu(X) < \infty$, we have $\mu(E_1) < \infty$, hence by continuity from above $\lim_{n \rightarrow \infty} \mu(E_n(k)) = 0$. Now for each k , choose n_k such that:

$$\mu(E_{n_k}(k)) < \frac{\epsilon}{2^k}.$$

Define $E := A^c \cup \bigcup_{k=1}^{\infty} E_{n_k}(k)$. We can see:

$$\begin{aligned} \mu(E) &= \mu\left(A^c \cup \bigcup_{k=1}^{\infty} E_{n_k}(k)\right) \\ &\leq \mu(A^c) + \mu\left(\bigcup_{k=1}^{\infty} E_{n_k}(k)\right) \\ &\leq 0 + \sum_{k=1}^{\infty} \mu(E_{n_k}(k)) \\ &< \sum_{k=1}^{\infty} \frac{\epsilon}{2^k} \\ &= \epsilon. \end{aligned}$$

Moreover, for all $x \in E^c$

□

§ 2.5. Integration in $\mathbb{R}^n$

missed a lot of stuff

Chapter 3

Differentiation

§ 3.1. Signed Measures

Definition 3.1.1. Let $(X, \mathcal{M})$ be a measurable space. The map $\nu : \mathcal{M} \rightarrow [-\infty, \infty]$ is a signed measure if:

- (1) $\nu(\emptyset) = 0$,
- (2) $\{\infty, \infty\} \not\subset \text{im } \nu$,
- (3) If $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}$ is pairwise disjoint, and $\sum_{n=1}^{\infty} \nu(E_n)$ is absolutely convergent whenever $\nu(\bigcup_{i=1}^{\infty} E_i)$ is finite, then $\nu(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \nu(E_i)$

Example 3.1.1.

- (1) If μ_1 and μ_2 are measures on $(X, \mathcal{M})$ and at least one is finite, then $\nu := \mu_1 - \mu_2$ is a signed measure.
- (2) Let $f : X \rightarrow [-\infty, \infty]$ be measurable, μ a measure, and either $\int f^+ d\mu$ or $\int f^- d\mu$ be finite. Then $\nu(E) := \int_E f d\mu$ is a signed measure.

Definition 3.1.2. Let ν be a signed measure on $(X, \mathcal{M})$ and $E \in \mathcal{M}$.

- (1) E is positive if $\nu(F) \geq 0$ for all $F \in \mathcal{M}$ with $F \subset E$.
- (2) E is negative if $\nu(F) \leq 0$ for all $F \in \mathcal{M}$ with $F \subset E$.
- (3) E is null if $\nu(F) = 0$ for all $F \in \mathcal{M}$ with $F \subset E$.

Proposition 3.1.1.

- (1) If $\{E_n\}_{n=1}^{\infty} \subset \mathcal{M}$ such that $E_1 \subset E_2 \subset E_3 \subset \dots$, then $\nu(\bigcup_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \nu(E_n)$.
- (2) If $\{E_n\}_{n=1}^{\infty} \subset \mathcal{M}$ such that $E_1 \supset E_2 \supset E_3 \supset \dots$, then $\nu(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \nu(E_n)$.
- (3) Any measurable subset of a positive (resp. negative, null) set is positive (resp. negative, null).
- (4) The union of countably many positive (resp. negative, null) sets is positive (resp. negative, null).
- (5) If $E \in \mathcal{M}$ is positive (resp. negative, null) and $F \subset E$, then $\nu(F) \leq \nu(E)$ (resp. $\nu(F) \geq \nu(E)$, $\nu(F) = 0$).

Theorem 3.1.1 (Hahn Decomposition Theorem). Let ν be a signed measure on $(X, \mathcal{M})$. There exists a positive set P and a negative set N such that:

- (1) $P \cup N = X$;
- (2) $P \cap N = \emptyset$;
- (3) If P' and N' are another such pair satisfying (1) and (2), then $P \Delta P'$ and $N \Delta N'$ are null.

Proof. Without loss of generality, suppose that ν does not take the value $+\infty$. Define:

$$m := \sup\{\nu(P) \mid P \in \mathcal{M}, P \text{ is positive}\}.$$

Find a sequence of positive sets $\{P_n\}_{n=1}^\infty$ such that $\nu(P_n) \rightarrow m$. Define $P := \bigcup_{n=1}^\infty P_n$. Then P is positive by Proposition ?? . Since $\nu(P_n) \leq \nu(P) \leq m$, by the squeeze theorem $\nu(P) = m$; i.e., $m < \infty$.
Claim: The set $N := X \setminus P$ is negative.

First, note that N cannot contain a positive set of positive measure. Indeed, if $P' \subset N$ where P' is positive and $\nu(P') > 0$, then $P \cup P'$ is positive and $\nu(P \cup P') = \nu(P) + \nu(P') > m$, which is a contradiction.

Second, if there exists a set $A \in \mathcal{M}$ such that $A \subset N$ and $\nu(A) > 0$, then there exists another subset $B \subset N$ such that $\nu(B) > \nu(A)$. Indeed, we have just shown that A cannot be positive, hence there exists a subset $C \subset A$ such that $\nu(C) < 0$. Therefore, $\nu(A \setminus C) = \nu(A) - \nu(C) > \nu(A)$. $\square$

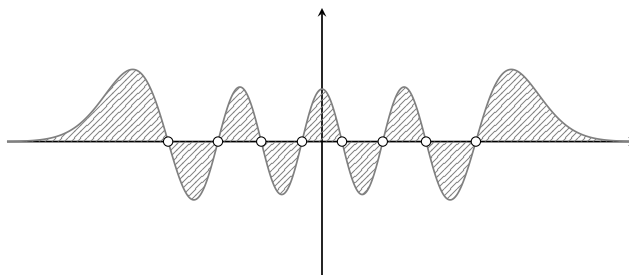


Figure 3.1: On $(\mathbb{R}, \mathcal{L})$ let m be Lebesgue measure and define the signed measure $\nu(E) = \int_E f \, dm$. The shading indicates the positive ($f \geq 0$) and negative ($f \leq 0$) contributions of f . By part (3) of the Hahn Decomposition Theorem, the decomposition $\mathbb{R} = P \cup N$ is unique up to null sets. Indeed, taking $P = \{x \mid f(x) \geq 0\}$, $P' = \{x \mid f(x) > 0\}$, $N = \{x \mid f(x) < 0\}$, and $N' = \{x \mid f(x) \leq 0\}$, we have that the sets $P \Delta P'$ and $N \Delta N'$ are null.